

Nucleosynthesis in Stars and the Chemical Enrichment of Galaxies

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Annu. Rev. Astron. Astrophys. 2013. 51:457–509

First published online as a Review in Advance on
July 3, 2013

The *Annual Review of Astronomy and Astrophysics* is
online at astro.annualreviews.org

This article's doi:
10.1146/annurev-astro-082812-140956

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Keywords

first star, galactic archaeology, gamma-ray burst, hypernova, metal-poor
star, supernova

Abstract

After the Big Bang, production of heavy elements in the early Universe takes place starting from the formation of the first stars, their evolution, and explosion. The first supernova explosions have strong dynamical, thermal, and chemical feedback on the formation of subsequent stars and evolution of galaxies. However, the nature of the Universe's first stars and supernova explosions has not been well clarified. The signature of the nucleosynthesis yields of the first stars can be seen in the elemental abundance patterns observed in extremely metal-poor stars. Interestingly, those patterns show some peculiarities relative to the solar abundance pattern, which should provide important clues to understanding the nature of early generations of stars. We thus review the recent results of the nucleosynthesis yields of mainly massive stars for a wide range of stellar masses, metallicities, and explosion energies. We also provide yields tables and examine how those yields are affected by some hydrodynamical effects during supernova explosions, namely, explosion energies from those of hypernovae to faint supernovae, mixing and fallback of processed materials, asphericity, etc. Those parameters in the supernova nucleosynthesis models are constrained from observational data of supernovae and metal-poor stars. Nucleosynthesis yields are then applied to the chemical evolution model of our Galaxy and other types of galaxies to discuss how the chemical enrichment process occurred during evolution.

1. INTRODUCTION

Just after the Big Bang, a cosmic primordial gas consisted mostly of H, He, and a small amount of light elements (Li, Be, B, etc.). The first heavier elements, such as C, O, Ne, Mg, Si, and Fe, must have synthesized in the evolution and explosion of the first stars (metal-free = Population III = Pop III stars) early in the history of the Universe. The massive first stars evolved to explode as the first supernovae, which released large explosion energies and ejected nucleosynthetically enriched materials. As a result, these first supernovae had strong dynamical, thermal, and chemical influences on the evolution of interstellar matter (supernova feedback) in the formation of the first galaxies. Therefore, identifying the first stars is crucial for understanding the early evolution of the Universe.

The formation of the first stars has been studied in theoretical detail (e.g., Ferrara 1998; Abel, Bryan & Norman 2000; Bromm & Yoshida 2011), but their typical masses [or initial mass function (IMF)] have not been clarified yet. If the first stars were more massive than $\sim 140 M_{\odot}$, the first stars evolved to become pair-instability supernovae (PISNe) (e.g., Heger & Woosley 2010) or to form intermediate-mass black holes. By contrast, if the first stars were less massive, they underwent core collapse. Nucleosynthesis yields and explosion energies are quite different between these two types of supernovae.

In the early Universe, when the metal content was extremely low, enrichment by a single supernova could dominate preexisting metal contents (e.g., Audouse & Silk 1995; Ryan, Norris & Beers 1996). As such, the abundance pattern of the enriched gas may reflect nucleosynthesis in the individual supernova. The next generation of stars formed from the enriched gas, and the long-lived low-mass stars may be observed as extremely metal-poor (EMP) stars (Beers & Christlieb 2005). Thus, the abundance patterns of EMP stars could constrain the nucleosynthetic yields of the Pop III supernovae and the mass range of the first stars. Beers & Christlieb (2005) have defined metal-poor stars with metallicity $[\text{Fe}/\text{H}]$ ($[\text{A}/\text{B}] = \log_{10}(N_{\text{A}}/N_{\text{B}}) - \log_{10}(N_{\text{A}}/N_{\text{B}})_{\odot}$, where the subscript $\odot$ refers to the solar value and N_{A} and N_{B} are the abundances of elements A and B, respectively) as follows: very metal-poor (VMP) stars for $-3 \leq [\text{Fe}/\text{H}] < -2$, EMP stars for $-4 \leq [\text{Fe}/\text{H}] < -3$, ultra metal-poor (UMP) stars for $-5 \leq [\text{Fe}/\text{H}] < -4$, hyper metal-poor (HMP) stars for $-6 \leq [\text{Fe}/\text{H}] < -5$, and mega metal-poor (MMP) stars for $[\text{Fe}/\text{H}] < -6$.

However, according to recent observations, the extremely unusual abundance patterns of several EMP stars, such as carbon-enhanced metal-poor (CEMP) and HMP stars (e.g., Beers & Christlieb 2005), are significantly different from previously known nucleosynthesis yields of massive stars. Nucleosynthesis in massive stars has been studied ever since the pioneering work by Hoyle (1946), Burbidge et al. (1957), Cameron (1957), Hoyle & Fowler (1960), and Hayashi, Hoshi & Sugimoto (1962) (for reviews, see, e.g., Arnett 1973, 1995; Trimble 1975). These new observations of EMP/CEMP/HMP stars raise important challenges to the stellar evolution and nucleosynthesis theory.

Another challenge to the conventional stellar evolution and supernova models is the establishment of the gamma-ray burst (GRB)-supernova connection (e.g., Woosley & Bloom 2006). To date, four GRB-associated supernovae have been confirmed spectroscopically. All are very energetic supernovae whose kinetic energy E exceeds 10^{52} erg, more than 10 times the kinetic energy of normal core-collapse supernovae. In this review, we use the explosion energy E to indicate the final kinetic energy of the explosion. We also use the term “hypernova” to describe a hyperenergetic supernova with $E_{51} = E/10^{51}$ erg $\gtrsim 10$ (Nomoto et al. 2004, 2006). The observed diversity of supernovae further extends to “faint supernovae” and superluminous supernovae (Quimby et al. 2007). These new types of supernova explosions may produce interesting nucleosynthesis yields, which can be tested with the observed elemental abundances in metal-poor stars.

On a larger scale, the chemical evolution of galaxies has been a powerful approach to studying galaxy formation (Tinsley 1980, Pagel 1997, Matteucci 2001). Galactic chemical evolution (GCE), in particular, has provided stringent constraints on stellar yields (Timmes, Woosley & Weaver 1995; Kobayashi et al. 2006). Isotope ratios have prompted new insights into the study of the details of stellar evolution and GCE as well as supernovae (e.g., Kobayashi, Karakas & Umeda 2011). We summarize the application of GCE and some topics of galaxy formation in Section 7. Recently, more detailed chemodynamical simulations have become available, providing kinematical and chemical properties of stars that can be directly compared with large-scale surveys (e.g., Kobayashi & Nakasato 2011). We do not discuss these here but note that stellar yields and the progenitor model of Type Ia supernovae compose the most important input physics.

In this article, motivated by these challenges, we review the recent results of the nucleosynthesis yields of mainly massive stars. Several groups have obtained stellar yields for a wide range of stellar masses and metallicities, which have been applied to studies of the chemical enrichment in the Universe. The comparison between such stellar nucleosynthesis yields and the abundance patterns of EMP/UMP/HMP stars can provide a new approach to determining the individual supernova mechanism, especially for Pop III supernovae.

The final stages of massive star evolution, supernova properties, and supernova chemical yields depend primarily on the progenitor's main-sequence masses M (e.g., Arnett 1996, Smartt 2009). In Section 2, we briefly discuss the fate and nucleosynthesis of asymptotic giant branch (AGB) and super AGB stars, including Type 1.5 supernovae and electron-capture supernovae. In Section 3, we describe general processes of nucleosynthesis in core-collapse supernovae and how the progenitor mass M and explosion energy E are estimated from the observations of supernovae/hypernovae. In Sections 4 and 5, we summarize the final stages of stellar evolution, supernova models, and nucleosynthesis yields as a function of stellar mass, metallicity, and explosion energy. In Section 6, we discuss the still-debated progenitors of Type Ia supernovae. In Sections 7 and 8, we apply the stellar yields to the modeling of the GCE and to the abundance profiling in individual metal-poor stars, respectively.

The nucleosynthesis table we compiled for GCE models is available online (Yields Table 2013). This table lists the ejected mass m_i of each isotope in units of solar mass that is a newly produced metal (processed metal) up to ^{74}Ge (see Section 4 for more details). This table does not include the initial metals that existed in the progenitor stars (unprocessed metals) that have been lost by stellar winds ($M_{\text{initial}} - M_{\text{final}}$). The associated abundance pattern is not scaled with the solar abundance. Instead, the pattern reflects the Galaxy at the time when the stars formed. The mass of the ejecta is given as $M_{\text{ejecta}} \equiv \sum_{i=1}^{74\text{Ge}} m_i = M_{\text{final}} - M_{\text{cut}}$, where M_{cut} is the remnant mass, i.e., the mass of the white dwarf, neutron star, or black hole.

2. EVOLUTION AND NUCLEOSYNTHESIS OF ASYMPTOTIC GIANT BRANCH AND SUPER ASYMPTOTIC GIANT BRANCH STARS

2.1. Yields Table

Stars in the mass range of $\sim 0.5 M_{\odot} - M_{\text{up,C}}$ form electron-degenerate C+O cores to become AGB stars. The upper limit of AGB stars, $M_{\text{up,C}}$, defined here as the minimum mass for the off-center C ignition to occur, is smaller for lower metallicity (Umeda et al. 1999; Gil-Pons, Gutiérrez & García-Berro 2007; Siess 2007). The yields table (Yields Table 2013) adopts $M_{\text{up,C}} = 3.5, 6.5, 6.5$, and 6.5 as well as $7 M_{\odot}$ for $Z = 0, 0.0001, 0.004, 0.008$, and 0.02 , respectively.

After the second dredge-up, these stars undergo thermal pulses of He shell burning and mass loss to form a planetary nebula and a C+O white dwarf. Such evolution and nucleosynthesis of AGB

stars have been reviewed by Busso, Gallino & Wasserburg (1999), Herwig (2005), and Käppeler et al. (2011). The resulting AGB yields are defined as the difference between the amount of the species in the wind and the initial amount in the progenitor star. Therefore, the yields of some isotopes (e.g., ^{15}N) are negative because they are destroyed during stellar evolution. However, in chemical evolution models, the abundances of such elements at time t may be lower than the adopted initial abundances, which causes numerical problems. This is often the case for ^{15}N , so we set the ^{15}N yield to 0 if it is negative.

For the yields table (Yields Table 2013), we take the AGB yields for $M = 1.0, 1.25, 1.5, 1.75, 1.9, 2.0, 2.25, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0$, and $6.5 M_{\odot}$ as well as $Z = 0.0001, 0.004, 0.008$, and 0.02 from Karakas (2010). (For $Z = 0.004$ and 0.008 , $2.0\text{-}M_{\odot}$ yields are derived from the interpolation between $1.9\text{-}M_{\odot}$ and $2.1\text{-}M_{\odot}$ models; the $6.5\text{-}M_{\odot}$ model is available only for $Z = 0.02$.) We use the yields and remnant mass for $Z = 0.0001$ at $0.0001 \leq Z \leq 0.001$ because the AGB yields do not vary greatly at this metallicity range. We use the yields and remnant mass for $Z = 0.02$ at $Z \geq 0.02$. The yields of the radioactive isotopes ^{26}Al and ^{60}Fe are added to ^{26}Mg and ^{60}Ni yields, respectively.

For $Z = 0$, theoretical models of stars undergo violent evolutionary episodes not seen at higher metallicities. The ingestion of hydrogen leads to an H flash, followed by a “normal” He shell burning phase. We take the yields and remnant masses from Campbell & Lattanzio (2008) for $M = 0.85, 1.0, 2.0$, and $3.0 M_{\odot}$. For Na, we assume that the yield is reduced by a factor of 10 because the old reaction rates were adopted in the calculations (Kobayashi, Karakas & Umeda 2011). At $M > 3.5 M_{\odot}$, no metals are produced. This assumption may not be valid but does not affect the average chemical evolution of galaxies.

2.2. C+O White Dwarfs versus Type 1.5 Supernovae

For solar metallicity, the C+O cores in AGB stars are not expected to have reached the Chandrasekhar (Chandra) mass mainly because wind mass loss prevents the C+O cores from growing their masses (e.g., Herwig 2005). Recently, a few full AGB calculations have appeared including a number of thermal pulses until the end of the AGB phase (e.g., Weiss & Ferguson 2009, Karakas 2010). The resultant C+O core mass depends strongly on the adopted mass-loss law, extra mixing (overshooting of convection), hot bottom burning, etc. Different laws give different results.

The mass-loss rates may not be as high as those observed in metallicities of $[\text{Fe}/\text{H}] \gtrsim -1$. A superwind also does not blow in lower metallicity environments (Zijlstra 2004). These changes of behavior occur at $[\text{Fe}/\text{H}] \approx -1$ (Bowen & Willson 1991). In contrast to the C+O cores in higher Z ($[\text{Fe}/\text{H}] > -1$) environments, those in lower environments may reach the Chandra mass, resulting in Type 1.5 supernovae (Iben & Renzini 1983), i.e., thermonuclear explosions of degenerate C+O cores in AGB stars. Because Type 1.5 supernovae produce as much ^{56}Ni as do Type Ia supernovae, whether Type 1.5 supernovae yields affect $[\alpha/\text{Fe}]$ for our Galaxy or dwarf galaxies is very sensitive to the mass range of Type 1.5 supernovae. Here, α denotes the so-called α -elements ranging from C to Ca. In addition, Type 1.5 supernovae generally refer to the Chandra mass explosions, thus producing Fe-peak elements with different ratios, e.g., $^{58}\text{Ni}/^{56}\text{Fe}$, from the sub-Chandra mass models for Type Ia supernovae.

2.3. Super Asymptotic Giant Branch Stars, O+Ne+Mg White Dwarfs, and Electron-Capture Supernovae

For stars in the mass range of $M_{\text{up,C}} < M \lesssim 10 M_{\odot}$, electrons are partially degenerate in a C+O core. In such a semidegenerate C+O core, neutrino cooling leads to off-center ignition of C when the C+O core mass exceeds the critical mass of $1.06 M_{\odot}$. The off-center C-burning shell

moves inward all the way to the center as a result of heat conduction (Nomoto 1984, Timmes & Woosley 1992).

2.3.1. Formation of O+Ne+Mg white dwarfs. After C is exhausted in the central region, an O+Ne+Mg core forms. The core mass does not exceed the critical mass of $1.37 M_{\odot}$ for neon ignition; hence, Ne burning is never ignited (Nomoto 1984). Subsequently, the O+Ne+Mg core becomes strongly degenerate, and the envelope becomes similar to that of super AGB stars (Eldridge & Tout 2004, Poelarends et al. 2008) with a thin He-burning shell that undergoes thermal pulses and s-process nucleosynthesis.

The final fate of these stars depends on the competition between the mass loss that reduces the envelope mass and the increase in the core mass through the H-He shell burning. If the mass loss is fast, an O+Ne+Mg white dwarf is formed, which could be the case for stars in the mass range of $M_{\text{up,C}} < M < M_{\text{up,Ne}}$. Here, $M_{\text{up,Ne}} \sim 9 \pm 1 M_{\odot}$ is the upper mass limit of the progenitor of an O+Ne+Mg white dwarf and is smaller for lower metallicities (Siess 2007, 2010; Poelarends et al. 2008; Pumo et al. 2009; Langer 2012).

2.3.2. Electron-capture supernovae. Stars in the mass range of $M_{\text{up,Ne}} < M < 10 M_{\odot}$ become electron-capture supernovae. In these stars, the core mass grows to $1.38 M_{\odot}$ and the central density reaches $4 \times 10^9 \text{ g cm}^{-3}$. The electron Fermi energy exceeds the threshold for electron captures $^{24}\text{Mg}(e^-, \nu) ^{24}\text{Na}(e^-, \nu) ^{20}\text{Ne}$ and $^{20}\text{Ne}(e^-, \nu) ^{20}\text{F}(e^-, \nu) ^{20}\text{O}$. The resultant decrease in Y_e triggers collapse (Miyaji et al. 1980; Nomoto 1987). In addition, the resultant explosion is induced by neutrino heating and is weak with kinetic energy as low as $E \sim 10^{50} \text{ erg}$ (Kitaura, Janka & Hillebrandt 2006). These stars produce little α -elements and Fe-peak elements but may be other sources of Zn and light p-nuclei.

Nucleosynthesis in the supernova explosion of a $9\text{-}M_{\odot}$ star has been presented by Wanajo et al. (2009) and Wanajo, Janka & Müller (2011). The largest overproduction is shared by ^{60}Zn , ^{70}Se , and ^{78}Kr . The ^{64}Zn production provides an upper limit to the occurrence of exploding O-Ne-Mg cores at approximately 20% of all core-collapse supernovae. This supernova may produce a significant amount of ^{48}Ca from a neutron-rich blob (Wanajo, Janka & Müller 2013). The ejecta mass of ^{56}Ni is $0.002\text{--}0.004 M_{\odot}$, much smaller than the $\sim 0.1 M_{\odot}$ in more massive progenitors.

The light-curve models of electron-capture supernovae (Tominaga, Blinnikov & Nomoto 2013) show a short plateau and a faint tail. Assuming that the faint tail can be influenced by the spin-down luminosity of a newborn pulsar, the above model well explains the observed properties of SN 1054 that formed the Crab Nebula (Nomoto et al. 1982). Furthermore, the envelope of the AGB star is C enhanced (Nomoto 1984) so that dust may easily be formed to induce mass loss. If the explosion energy is low enough, the observed properties of faint supernovae of Type IIn, such as SN2008S and similar transients (Prieto et al. 2008), may also be explained with electron-capture supernovae.

As described above, stars with $M_{\text{up,Ne}} < M < 10 M_{\odot}$ may explode as electron-capture supernovae (Nomoto 1984; Kitaura, Janka & Hillebrandt 2006), but the metal production (lighter than Fe) is predicted to be very small. Furthermore, this mass range is narrower for more metal-rich environments. In the yields table (Yields Table 2013), we assume that no metals are produced from stars in the range of $M_{\text{up,C}} < M < 10 M_{\odot}$. The remnant mass at $M_{\text{up,Ne}}$ is extrapolated for $Z < 0.02$. The remnant mass M_{remnant} is set as 1.01, 1.12, and $1.15 M_{\odot}$ for 7, 8, and $10 M_{\odot}$, respectively.

3. EXPLOSIVE NUCLEOSYNTHESIS IN CORE-COLLAPSE SUPERNOVAE

Massive stars in the range of 10 to $\sim 140 M_{\odot}$ form Fe cores and undergo core collapse. The presupernova abundance distributions in the $20\text{-}M_{\odot}$ star models are shown in **Figure 1**. If a

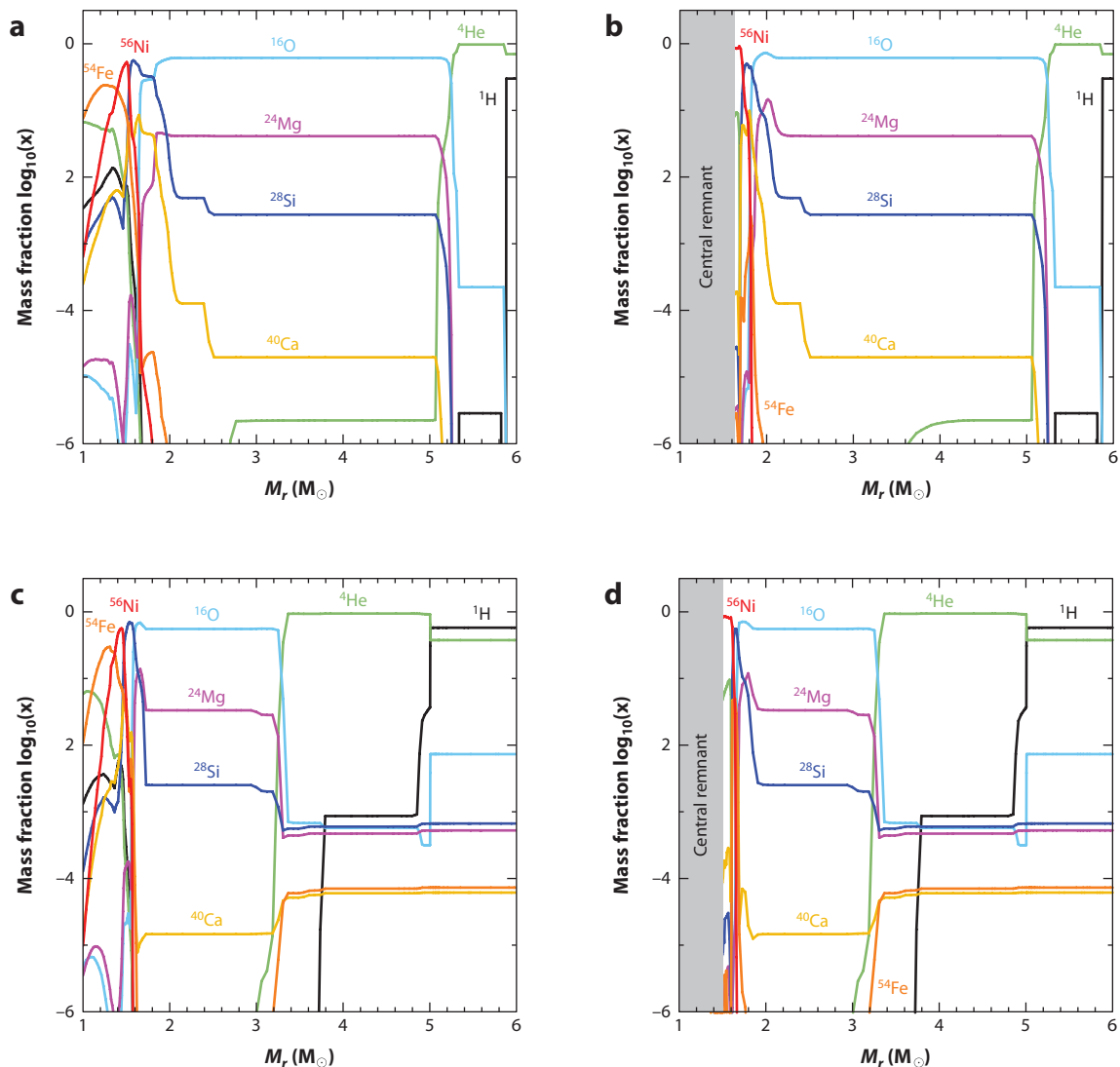


Figure 1

Abundance distribution against an enclosed mass M_r before (*a, c*) and after (*b, d*) an explosion of a Population (Pop) III $20\text{-}M_{\odot}$ star with $E_{51} = 1$ (*a, b*) and solar metallicity $20\text{-}M_{\odot}$ star with $E_{51} = 1$ (*c, d*). A Pop III star is more compact. Thus, compared with a solar metallicity star, each layer is more extended in mass. The ejected Fe is explosively synthesized in the Si and O layers with $Y_e \sim 0.5$ in the progenitor star.

collapse is successfully transformed into an explosion, stellar materials undergo shock heating and explosive nucleosynthesis. In “explosive nuclear burning,” the timescale of the main nuclear reaction is shorter than the hydrodynamical timescale of an expansion.

The mechanism of transformation from collapse to explosion is not fully understood (e.g., Janka 2012, Kotake et al. 2012, Bruenn et al. 2013, Burrows 2013). Thus, simulations of explosive nucleosynthesis usually need to generate a shock wave at a certain “mass cut” via the introduction of thermal or kinetic energy. The final kinetic energy of explosion E and the mass cut that separates the

ejecta from the collapsing core need to be constrained from the supernova observations. Abundance distributions after explosive nucleosynthesis are shown in **Figure 1**. Here, we describe the general features of explosive nucleosynthesis (e.g., Woosley & Weaver 1995; Arnett 1996; Thielemann, Nomoto & Hashimoto 1996; Boyd 2008) whose explosively produced elements depend on the initial abundances, the peak temperature T_p attained through the passage of the shock, the density ρ_p when the peak temperature is reached, and the number of electrons per nucleon (electron mole number) Y_e .

3.1. Neutron-Proton Ratio Near the Mass Cut

The number of electrons per nucleon Y_e in the deepest layer around the mass cut determines the isotopic ratio of Fe-peak elements in explosive nucleosynthesis. For smaller Y_e , more neutron-rich species are synthesized. The important processes that involve weak interactions and determine the distribution of Y_e are listed below. The first two processes occur during the quasi-static presupernova evolution, and the other processes occur during the hydrodynamical phases.

1. The initial C, N, and O elements are processed to ^{14}N through the CNO cycle of H burning and then processed into neutron-rich species ^{22}Ne through $^{14}\text{N}(\alpha, \gamma)^{18}\text{F}(\beta^+)^{18}\text{O}(\alpha, \gamma)^{22}\text{Ne}$ during He burning. Thus, the initial metallicity effects should appear in the abundances of some odd- Z elements, which are the products of He and C burning.
2. Neutronization due to electron capture and positron decay occurs during convective O and Si core and shell burning. This neutronization process dominates the effect of the initial metallicity. Overshooting at the edge of the convective shell also affects the values and distribution of Y_e . Three-dimensional (3D) simulations are important to determining the size and asymmetry of the convective layer around the Fe core (Arnett & Meakin 2011).
3. During core collapse, Y_e decreases mainly as a result of electron capture. Collapsing materials with very small Y_e may not be ejected except via jet-like ejection.
4. Following core bounce, a shock wave forms and propagates outward to eject some outer materials. In the shocked ejecta of 8–10- $M_\odot$ stars, high densities cause electron capture to produce moderately neutron-rich materials. In more massive stars, by contrast, the densities are too low for electron capture to reduce Y_e substantially.
5. The neutrino heating mechanism is used to generate successful explosions. In such models, neutrino absorption significantly increases Y_e , thereby forming proton-rich ejecta (e.g., Janka, Buras & Rampp 2003; Liebendörfer et al. 2003; Fröhlich et al. 2006).
6. The region where neutrino absorption and Y_e variation occur is Rayleigh-Taylor unstable. In multidimensional models, a large range of Y_e distribution is produced, which leads to interesting nucleosynthesis (e.g., Janka, Buras & Rampp 2003; Wanajo, Janka & Müller 2011).
7. The ejected mass of the neutron-rich layer also depends on the amount of fallback (discussed below). Y_e distribution, E , and the multidimensional effects are coupled with each other. Thus, self-consistent explosion models are needed.

3.2. Mixing and Fallback Model

The ejected masses of radioactive ^{56}Ni and neutron-rich Fe-peak elements depend critically on the location of the mass cut. The mixing and fallback model as first introduced by Umeda & Nomoto (2002) is an important tool for understanding how the mass cut is determined in the explosion. To begin, the inner materials are assumed to be mixed by Rayleigh-Taylor instabilities and/or aspherical explosions during shock wave propagation in the star (e.g., Hachisu et al. 1990;

Müller, Fryxell & Arnett 1991). Furthermore, the postshock materials are convectively unstable as a result of deceleration. Later, gravity causes some fraction of the materials in the mixing region to undergo fallback onto the central remnant (e.g., Woosley & Weaver 1995, Iwamoto et al. 2005), and the remaining material is ejected into interstellar space. The yields are made from materials that exist above the mixing region and do not fall back (e.g., Joggerst et al. 2010). The degree of fallback depends on the explosion energy, the gravitational potential, and asphericity (e.g., Fryer, Hungerford & Young 2007; Moriya et al. 2010b).

Fallback can occur in relatively low-energy explosions as well as in very energetic jet-like explosions. In fact, Maeda & Nomoto (2003), Tominaga et al. (2007), and Tominaga (2009) have simulated jet-like explosions and shown that their nucleosynthesis yields can be similar to those found in the mixing and fallback model because the materials around the jet axis are ejected and the materials around the equatorial plane fall back onto the central remnant. The jet-like explosion is consistent with a recently observed association between hypernovae and GRBs with highly collimated relativistic jets. The mixing and fallback model mimics such aspherical explosions, although the spherical model tends to require larger explosion energies than does the jet model to obtain similar yields (Maeda & Nomoto 2003).

3.3. Postshock Temperature and Explosive Nuclear Burning

Following shock passage, the region is radiation dominated, so the peak temperature is approximately related to the stellar radius r and the deposited energy E^* ($= E - E_{\text{grav}}$, where E_{grav} is the gravitational binding energy) by

$$r = 3.16 \cdot 10^4 (E_{51}^*)^{1/3} / T_9^{4/3} \text{ km}, \quad (1)$$

where E_{51}^* is the deposited energy in units of 10^{51} erg and $T_9 = T_p/10^9$ K is the peak temperature. Therefore, assuming $E_{51}^* = 1$, explosive nuclear burning is classified into four cases based on T_9 : (a) $T_9 = 5$ ($r = 3,700$ km), (b) $T_9 = 4$ ($r = 4,980$ km), (c) $T_9 = 3.3$ ($r = 6,430$ km), and (d) $T_9 = 2.0$ ($r = 11,800$ km). The relation between the presupernova radius r and the enclosed mass M_r depends on the progenitor's mass.

3.3.1. Complete Si burning. For $T_p > 5 \times 10^9$ K, explosive Si burning leads to nuclear statistical equilibrium for $\rho > 10^8$ g cm $^{-3}$. At lower densities of $\rho < 10^8$ g cm $^{-3}$, which are typical for core-collapse supernovae, α -rich freeze-out occurs. Some ^4He remains after the freeze-out without recombination into the Fe-peak elements. (a) For $Y_e \sim 0.50$, ^{56}Fe and ^{57}Fe are produced as radioactive decay products of ^{56}Ni and ^{57}Ni . These decays are important energy sources for supernova light curves. $^{56,57}\text{Fe}$ are also directly synthesized. (b) In more neutron-rich (smaller Y_e) regions, ^{58}Ni is directly synthesized, whereas $^{60,62}\text{Ni}$ are produced both directly and from radioactive decays of $^{60,62}\text{Zn} \rightarrow ^{60,62}\text{Cu} \rightarrow ^{60,62}\text{Ni}$. (c) ^{59}Co is the decay product of ^{59}Cu , and ^{63}Cu is directly synthesized. (d) ^{64}Zn , which is important in abundance profiling of metal-poor stars, is made as ^{64}Ge in an α -rich freeze-out in the innermost zones, i.e., at high T_p . (e) ^{51}V is another after-decay product found in the complete Si-burning region. (f) Radioactive decay of ^{44}Ti into ^{44}Ca is also important in supernova light curves at late times.

3.3.2. Incomplete Si burning. For 4×10^9 K $< T_p < 5 \times 10^9$ K, incomplete Si burning produces ^{56}Ni , ^{28}Si , ^{32}S , ^{36}Ar , ^{40}Ca , and ^{44}Ti . In the incomplete Si-burning region, the after-decay products include Cr and Mn, i.e., $^{52}\text{Fe} \rightarrow ^{52}\text{Cr}$ and $^{55}\text{Co} \rightarrow ^{55}\text{Mn}$.

3.3.3. Fe-peak elements and neutron excess. For abundance profiling of metal-poor stars, the ratios of Fe-peak elements to Fe with respect to the solar ratio ($[\text{V}/\text{Fe}]$, $[\text{Cr}/\text{Fe}]$, $[\text{Mn}/\text{Fe}]$, $[\text{Co}/\text{Fe}]$, $[\text{Ni}/\text{Fe}]$, $[\text{Cu}/\text{Fe}]$, and $[\text{Zn}/\text{Fe}]$) in the explosive Si-burning region are important. These ratios depend on Y_e as follows: For larger Y_e (~ 0.5), the amounts of $^{64}\text{Ge} \rightarrow ^{64}\text{Zn}$, $^{52}\text{Fe} \rightarrow ^{52}\text{Cr}$, $^{48}\text{Cr} \rightarrow ^{48}\text{Ti}$, and $^{44}\text{Ti} \rightarrow ^{44}\text{Ca}$ are also larger. In other words, the amount of decay products, such as ^{64}Zn , ^{52}Cr , and ^{48}Ti , tends to be larger for larger Y_e . In contrast, ^{58}Ni , ^{59}Co , and ^{63}Cu have large neutron excess, so that their productions (and, thus, $[\text{Ni}/\text{Fe}]$, $[\text{Co}/\text{Fe}]$, and $[\text{Cu}/\text{Fe}]$) tend to be smaller for larger Y_e . Mn is produced in the incomplete Si-burning region, where $[\text{Mn}/\text{Fe}]$ is smaller for larger Y_e . The ratio with respect to Fe is not trivial because ^{56}Ni production is larger for larger Y_e . For example, $[\text{Cr}/\text{Fe}]$ is almost constant for increasing Y_e because the amounts of both ^{52}Fe and ^{56}Ni increase in almost the same ratio.

3.3.4. Explosive O-Ne-C burning. For $3.3 \times 10^9 \text{ K} < T_p < 4.0 \times 10^9 \text{ K}$, explosive Ne burning produces ^{16}O , ^{28}Si , ^{32}S , and their isotopes. For $2.0 \times 10^9 \text{ K} < T_p < 3.3 \times 10^9 \text{ K}$, explosive C burning produces ^{20}Ne , ^{24}Mg , and their isotopes. However, the peak temperature is too low to change the initial abundances appreciably. For $T_p < 2 \times 10^9 \text{ K}$, no explosive burning occurs. For the O-rich layer with $2 \times 10^9 \text{ K} < T_p < 3 \times 10^9 \text{ K}$, where explosive Ne and C burning takes place, the p -process occurs via the photodisintegration of nuclei: (γ, p) , (γ, n) , (γ, α) . The inner part of the star for which $T_p > 2 \times 10^9 \text{ K}$ undergoes explosive nucleosynthesis according to the above classification. The outer part, including most of the original O-rich layer, is ejected relatively unprocessed by explosive nuclear burning. This leads to a mass dependence of the ratio between the explosive and hydrostatic burning products.

3.4. Explosion Energy

As mentioned in Section 1, the explosion energies of core-collapse supernovae are fundamentally important quantities, and an estimate of $E \sim 1 \times 10^{51} \text{ erg}$ has often been used for nucleosynthesis calculations. A good example is SN1987A in the Large Magellanic Cloud, whose energy is estimated to be $E = (1.0\text{--}1.5) \times 10^{51} \text{ erg}$ from its early light curve (Arnett et al. 1989, Shigeyama & Nomoto 1990, Blinnikov et al. 2000). Important changes have come from the establishment of the connection between long GRBs and core-collapse supernovae from GRB 980425/SN 1998bw (Galama et al. 1998), GRB 030329/SN 2003dh (Stanek et al. 2003, Hjorth et al. 2003), GRB 031203/SN 2003lw, and GRB 120422A/SN2012bz (Melandri et al. 2012). These GRB supernovae are of Type Ic, showing the broad-line spectra (supernovae BL-Ic). The properties of these GRB supernovae, such as the ejected mass (the main-sequence mass of the progenitor) and the kinetic energy of explosion, have been estimated from comparisons between observed light curves and spectra and associated theoretical models (Nomoto et al. 2006). As summarized in **Figure 2**, these GRB supernovae have similar properties: They are all hypernovae with $E_{51} \sim 30\text{--}50$ and synthesize $0.3\text{--}0.5 M_\odot$ of ^{56}Ni (Iwamoto et al. 1998; Woosley, Eastman & Schmidt 1999). The mass estimates, obtained from fitting the optical light curves and spectra, place hypernovae at the high-mass end of supernova progenitors. The masses of radioactive ^{56}Ni are estimated from comparisons between theoretical and observed light curves.

Hypervovae are also characterized by asphericity according to observations of polarization and emission-line features (e.g., Maeda et al. 2002, Wang et al. 2002). The explosion energy of aspherical models for hypernovae tends to be smaller than that of spherical models by a factor of 2–3, but it is still as high as $E_{51} \gtrsim 10$ (Maeda & Nomoto 2003). A connection between X-ray flash and supernovae has also been found: GRB 060218/SN 2006aj (Pian et al. 2006) and

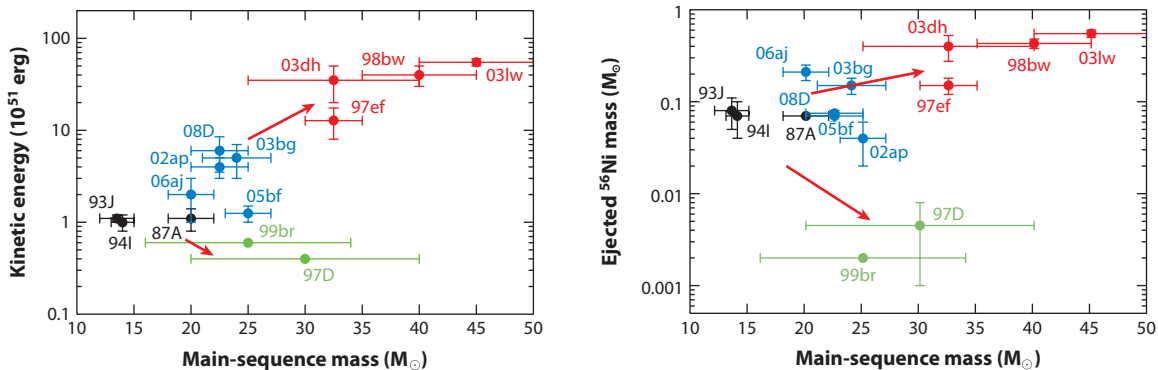


Figure 2

Explosion energy and ejected ^{56}Ni mass as a function of the main-sequence mass of the progenitors for several supernovae/hypernovae. Explosions of 13–25 $M_{\odot}$ stars cluster at normal supernovae, whereas explosions of 25–40 $M_{\odot}$ stars have a large variety ranging from hypernovae to faint supernovae.

GRB 100316D/SN 2010bh (Bufano et al. 2012). Compared with the above GRB supernovae, SN 2006aj is less energetic ($E_{51} \sim 2$) and its progenitor mass is smaller ($\sim 20 M_{\odot}$) (Mazzali et al. 2006), whereas SN 2010bf may be as energetic as $E_{51} \sim 10$ (Bufano et al. 2012).

Other non-GRB “hypernovae” have been recognized, such as SN 1997ef (Iwamoto et al. 2000) and SN 2002ap (Mazzali et al. 2002). These hypernovae span a wide range of properties, although they all appear to be highly energetic compared with normal core-collapse supernovae. In contrast, Type II supernovae 1997D and 1999br are very faint supernovae with very low E (Turatto et al. 1998, Hamuy 2003, Zampieri et al. 2003). As shown in **Figure 2**, E and the mass of ^{56}Ni ejected $M(^{56}\text{Ni})$ are a function of the main-sequence mass M of the progenitor star. Therefore, we propose that supernovae from stars with $M \gtrsim 20\text{--}25 M_{\odot}$ have different E and $M(^{56}\text{Ni})$, with a bright, energetic “hypernova branch” at one extreme and a faint, low-energy supernova branch at the other (Nomoto et al. 2003). For faint supernovae, the explosion energy was so small that most ^{56}Ni fell back onto the compact remnant. Thus, the faint supernova branch may become a “failed” supernova branch at larger M . Between the two branches, there may also be a variety of supernovae (Hamuy 2003).

This trend could be interpreted as follows: Stars more massive than $\sim 25 M_{\odot}$ form a black hole at the end of their evolution. Stars with nonrotating black holes are likely to collapse “quietly,” thereby ejecting a small amount of heavy elements (faint supernovae). In contrast, stars with rotating black holes are likely to give rise to hypernovae. Hypernova progenitors may form rapidly rotating cores when a companion star within a binary system spirals in.

4. CORE-COLLAPSE SUPERNOVAE FROM MASSIVE STARS

As discussed in Section 1, the evolution and final fates of Pop III stars were particularly important in the formation of the first galaxies and, thus, the early evolution of the Universe. The fates of Pop III stars depend on the mass to which the initially low-mass stars grow through mass accretion. Radiative feedback from growing stars can significantly reduce the mass-accretion rate (Hosokawa et al. 2011). However, this rate depends on the geometry and, thus, the angular momentum of accreting gas. For various cases of feedback and mass-accretion rates, Ohkubo et al. (2009) calculated the evolution of accreting Pop III stars to show that massive stars may

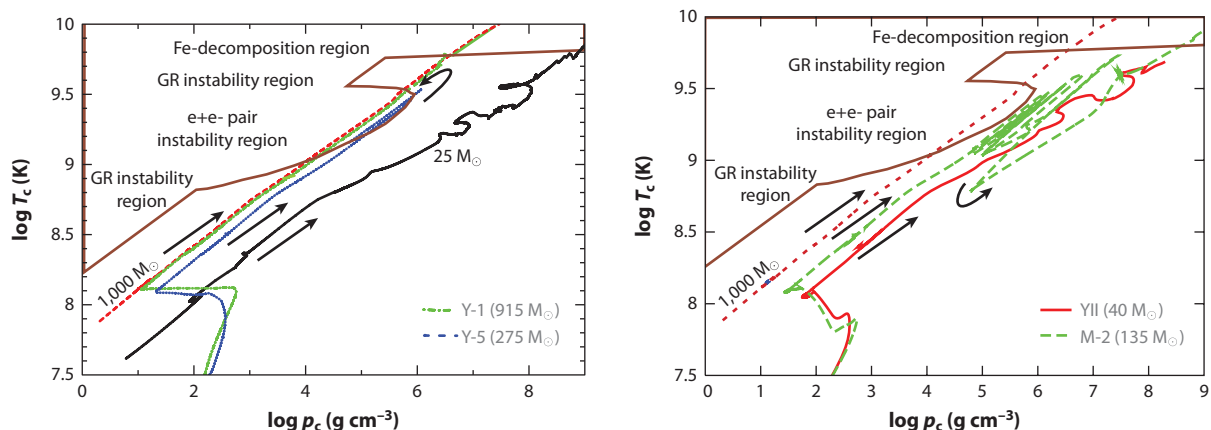


Figure 3

Evolutionary tracks of the central temperature and density of Pop III massive stars with mass accretion (Ohkubo et al. 2009). Arrows indicate the direction of evolution. Numbers in parentheses are the final masses for various accretion rates and radiative feedback effects. Evolutionary tracks without accretion are also shown for $M = 1,000$ and $25 M_{\odot}$ (Ohkubo et al. 2006). Abbreviation: GR, general relativity.

form if the mass accretion is not significantly reduced during main-sequence evolution (**Figure 3**) (Ohkubo et al. 2009). Pop III stars may be more massive than $\sim 300 M_{\odot}$, if rapid mass accretion continues during the whole main-sequence phase of Pop III stars (Ohkubo et al. 2006, 2009).

As shown in **Figure 3**, massive stars in the range of $M = 10\text{--}140 M_{\odot}$ undergo Fe core collapse at the end of their evolution and become Type II and Ib/c supernovae unless the entire star collapses into a black hole with no mass ejection. Here, the models of stellar evolution, supernovae explosions, and nucleosynthesis are described as a function of the main-sequence mass M . These models are constrained from the comparison of theoretical supernovae light curves and spectra with observations.

4.1. General Mass Dependence of Stellar Structure and Evolution

Evolution of stars with $M \gtrsim 10 M_{\odot}$ is different from that of less-massive stars because electrons do not become strongly degenerate owing to sufficiently high entropy in the central region of the core. By contrast, stars with $M < 10 M_{\odot}$ develop strongly degenerate He, C+O, and O+Ne+Mg cores as a result of the smaller entropy of the central regions. Accordingly, the pressure of massive stars is dominated by ideal gas and radiation pressures, resulting in strong temperature dependence. The core then has a negative specific heat and is gravothermally unstable. Central density and temperature increase as a result of entropy loss due to photon and neutrino energy losses (see equations 2 and 3 in Nomoto & Hashimoto 1988).

During core H and He burning, stellar evolution is driven by photon energy loss, and the mass of the convective core is larger for larger M . Thus, the resultant masses of the He and C+O cores are also larger for more massive stars (although the size of the convective burning shell is more complicated). By contrast, during core C burning and subsequent nuclear burning, evolution is driven by neutrino energy loss, and the loss rate is higher for higher temperatures. Accordingly, entropy loss occurs only in the small core with high temperature. The outer thermal structure

with high entropy is essentially frozen because the timescale of radiative energy transfer is longer than the evolutionary timescale determined by neutrino loss in the central region. During nuclear burning, the convective core is confined in the small central region. In other words, core size is determined by the balance between neutrino energy loss and nuclear energy release, thus becoming insensitive to M . This is the main reason the masses of Si and Fe cores are not very sensitive to M . Note again that the size of convective shell burning shows more complicated dependence on M , such that Fe core mass is not a monotonic function of M .

In the explosion, outer O-rich materials, which are mostly unprocessed materials, are ejected. Thus, the ejected O mass sharply increases with increasing M . By contrast, products of explosive O and Si burning, i.e., Si and Fe, are not sensitive to M , explaining why $[O/Fe]$ in the ejecta steeply increases with M .

4.2. Yields Tables for Core-Collapse Supernovae

Because $10\text{--}140 M_{\odot}$ stars make a key contribution to the enrichment of C- to Fe-peak elements, nucleosynthesis yields tables from stars in this mass range have been used in studies of cosmic energy and GCE. In particular, the yields table as a function of M and Z provided by Woosley & Weaver (1995) has been widely used in chemical evolution studies. However, the authors did not include mass loss in their presupernova evolution calculations. Another problem is that their core-collapse supernova models tend to produce more Fe than did SNe 1987A, 1993J, and 1994I (Section 4.4) by a factor of ~ 2 because of the relatively deep mass cut. Such large Fe yields lead to $[\alpha/Fe] \sim 0$ in the ejecta, which would not be consistent with $[\alpha/Fe] > 0.2$ observed in metal-poor stars. Another problem is the lack of room for Type Ia supernovae to add Fe in chemical evolution models. Thus, in some of these models (Timmes, Woosley & Weaver 1995; Romano et al. 2010), Fe yields are artificially reduced by a factor of 2. However, no reduction is made for other Fe-peak elements, which leads to wrong abundance ratios among the Fe-peak elements.

Portinari, Chiosi & Bressan (1998) obtained C+O core masses from stellar evolution models with mass loss and adopted Woosley & Weaver (1995) yields for those C+O core masses. This approach has the following problems: (a) The Fe yield problem in Woosley & Weaver (1995) remains. (b) The C+O core structure (and the resultant yields) retains the memory of mass loss. Thus, the yields are significantly different between models with and those without mass loss even if they have the same core mass (Woosley, Langer & Weaver 1993). (c) Mg production in the model by Woosley & Weaver (1995) with $M = 40 M_{\odot}$ and $E = 1 \times 10^{51}$ erg is unreasonably small compared with that in other models (Woosley & Weaver 1995; Nomoto et al. 1997a, 2006). Timmes, Woosley & Weaver (1995) do not use this model, but Portinari, Chiosi & Bressan (1998) overly apply it in their GCE models. The yields table for the solar metallicity models by Hashimoto, Nomoto & Shigeyama (1989); Thielemann, Nomoto & Hashimoto (1996); and Nomoto et al. (1997a) were used to determine the mass cut by applying the ^{56}Ni mass versus the main-sequence mass relation obtained from supernova light curves and spectra, but no mass loss was included.

Several groups have provided recent versions of nucleosynthesis yields for core-collapse supernovae and hypernovae (e.g., Limongi & Chieffi 2006, 2012; Nomoto et al. 2006; Tominaga, Umeda & Nomoto 2007; Heger & Woosley 2010). The yields we describe are given in Yields Table (2013) as functions of the main-sequence mass of the progenitor M and metallicity Z , including mass loss. Yields of Fe-peak elements are self-consistently obtained from mixing and fallback. The effect of rotation (e.g., Maeder 1992, 2009; Hirschi 2007) is not included. These yields are taken from Nomoto et al. (2006); Kobayashi et al. (2006); the three updated models of Kobayashi, Karakas & Umeda (2011); and unpublished $Z = 0.05$ models (N. Tominaga,

N. Iwamoto, N. Nomoto, submitted). In view of the uncertainty in Y_e , most precollapse models adopt $Y_e = 0.4997$ in the incomplete Si-burning region.

For normal core-collapse supernovae, $E_{51} = 1$ is assumed. The supernova table for $Z = 0$ gives yields for $M = 11, 13, 15, 18, 20, 25, 30, 40$, and $100 M_\odot$. The supernova tables for $Z = 0.001, 0.004, 0.008, 0.02$, and 0.05 give yields for $M = 13, 15, 18, 20, 25, 30$, and $40 M_\odot$. ($Z = 0.008$ yields are interpolated with yields between $Z = 0.004$ and $Z = 0.02$.) For hypernovae, the (M – E) relation as estimated from observations and models of supernovae (**Figure 2**) is adopted. Thus, the hypernova table for $Z = 0$ gives yields for a set of $(M/M_\odot, E_{51}) = (20, 10), (25, 10), (30, 20), (40, 30)$, and $(100, 60)$. The hypernova table for $Z = 0.001, 0.004, 0.008, 0.02$, and 0.05 give yields for a set of $(M/M_\odot, E_{51}) = (20, 10), (25, 10), (30, 20)$, and $(40, 30)$. Tables of radioactive species as in Tominaga, Umeda & Nomoto (2007) are also found in Yields Table (2013). Based on yields taken from that table (Yields Table 2013), **Figure 4** shows remnant and ejected element masses as a function of the progenitor mass for $Z = 0$ and 0.02 .

4.3. 10–13- $M_\odot$ Stars and Faint Supernovae

Efficient neutrino cooling in the semidegenerate O+Ne+Mg core causes 10–13- $M_\odot$ stars to undergo off-center Ne ignition (Nomoto & Hashimoto 1988). By contrast, Ne is ignited at the center in the 13- $M_\odot$ star. The Ne flame propagates inward as a result of core contraction. For 10–12- $M_\odot$ stars, the fate depends on whether the Ne flame reaches the center. If the Ne flame is quenched, the central region could form an electron-degenerate O-Ne-Mg core of 1.34–1.37 $M_\odot$ surrounded by a layer with Ne-burning products, i.e., an Si-Fe-rich layer. The star would then become an electron-capture supernova as in lower-mass stars, although further study is necessary.

For stars with $M \gtrsim 12 M_\odot$, the Ne flame reaches the center (Nomoto & Hashimoto 1988). In this case, the core evolves into an Fe core smaller than 1.4 $M_\odot$, as described in Müller, Janka & Heger (2012). The explosion of a star with such a small-mass Fe core could be powered by neutrino heating, thereby becoming a weak supernova as found in electron-capture supernovae with O-Ne-Mg cores (Müller, Janka & Heger 2012). Such a weak supernova would eject only a small amount of heavy elements.

The Fe core collapse of these stars would certainly lead to the formation of a neutron star because there exists a steep density gradient around $\sim 1.4 M_\odot$, and the outer envelope is too extended to accrete and further increase the mass of the collapsing core beyond 1.4 $M_\odot$. The resultant supernovae tend to be faint because a negligibly small amount of ^{56}Ni is ejected (Müller, Janka & Heger 2012). Such a supernova may correspond to faint supernova (Smartt 2009). Type Ib SN 2005cz, which is unusually faint and rapidly fading, may be an example of such a faint supernova (Kawabata et al. 2010). Its O emission lines in late-time spectra are much weaker than those of Ca, contrary to most other core-collapse supernovae. Kawabata et al. (2010) explain these unusual features of SN 2005cz in terms of an He core supernova from the low-mass end of the core-collapse progenitors (i.e., either ~ 8 –10 or 10–12 $M_\odot$) in close binaries.

Another candidate fallback supernova is the peculiar SN 2008ha (Valenti et al. 2009, Foley et al. 2009), which is very faint and whose explosion energy is estimated to be very small. Such an explosion can be modeled as fallback on the core-collapse supernova. As Moriya et al. (2010b) have shown, explosion of the C+O core of the 13- $M_\odot$ star can account for the properties of SN 2008ha. In their model, kinetic energy, ejecta mass, and the ejected ^{56}Ni mass are 1.4×10^{48} erg, 0.08 $M_\odot$, and 0.003 $M_\odot$, respectively. However, SN 2008a shows a prominent Si feature in the earliest spectrum and, thus, may be a very peculiar Type Ia supernova (Foley et al. 2010). The

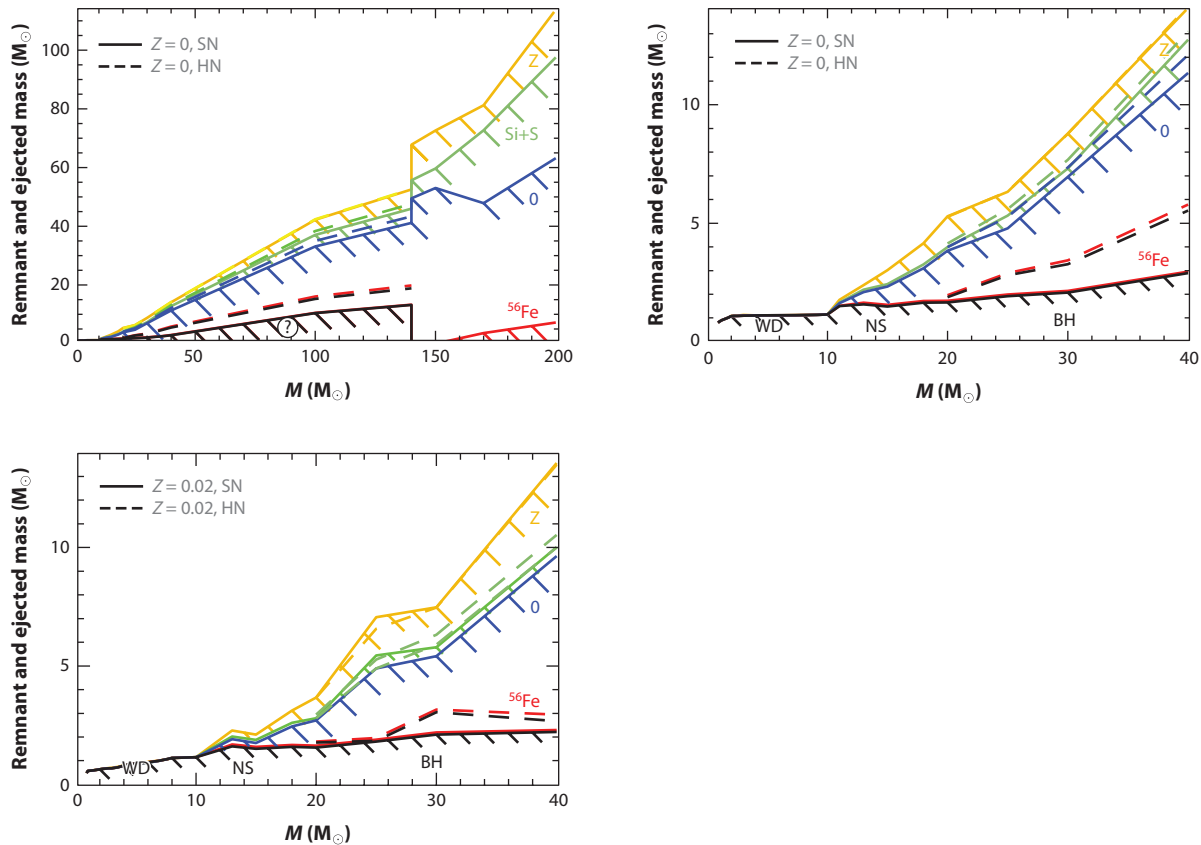


Figure 4

Remnant and ejected element masses as a function of the progenitor mass with $Z = 0$ and $Z_{\odot}$. Yields are taken from Yields Table (2013). The black, red, blue, green, and yellow lines indicate M_{rem} , $M(^{56}\text{Fe}) + M_{\text{rem}}$, $M(\text{O}) + M(^{56}\text{Fe}) + M_{\text{rem}}$, $M(\text{Si} + \text{S}) + M(\text{O}) + M(^{56}\text{Fe}) + M_{\text{rem}}$, and $M(\text{Z}) + M_{\text{rem}}$. Solid and dashed lines represent supernovae and hypernovae, respectively. Black lines represent the mass of remnants M_{rem} : white dwarfs (WD) for $M \lesssim 8 M_{\odot}$, neutron stars (NS) for $M \sim 8\text{--}20 M_{\odot}$, and black holes (BH) for $M \sim 20\text{--}140 M_{\odot}$. Above $\sim 50 M_{\odot}$, the remnant mass is uncertain. $\sim 140\text{--}300 M_{\odot}$ stars do not leave any remnants. Red lines indicate the summation of Fe and remnant mass, i.e., $M(^{56}\text{Fe}) + M_{\text{rem}}$. Blue and green lines indicate $M(\text{O}) + M(^{56}\text{Fe}) + M_{\text{rem}}$ and $M(\text{Si} + \text{S}) + M(\text{O}) + M(^{56}\text{Fe}) + M_{\text{rem}}$, respectively. Yellow lines indicate the summation of total metal and remnant masses, i.e., $M(\text{Z}) + M_{\text{rem}}$. Red-line slopes are steeper than those of the blue lines at $13\text{--}50 M_{\odot}$, which indicates the mass dependence of $[\text{O}/\text{Fe}]$ on the progenitor masses of Type II supernovae and hypernovae. The difference between the red and green lines is large at $> 140 M_{\odot}$, which means pair-instability supernovae generate much larger $[(\text{Si}, \text{S})/\text{O}]$ ratios than do Type II supernovae and hypernovae. Metal production slightly decreases at high metallicity.

final fate of $10\text{--}13 M_{\odot}$ stars needs to be studied further, but supernovae arising from this mass range would not contribute much to GCE.

4.4. $13\text{--}25 M_{\odot}$ Stars and Normal Supernovae

Following Fe core collapse, $13\text{--}25 M_{\odot}$ stars form a neutron star and produce significant amounts of heavy elements ranging from α -elements to Fe-peak elements. The boundary mass between neutron star and black hole formation, $M_{\text{NS/BH}} \sim 20\text{--}25 M_{\odot}$, is only tentative. For this range

of stars, three groups have presented yields tables noting various metallicities including Pop III (Nomoto & Hashimoto 1988; Woosley & Weaver 1995; Heger & Woosley 2002, 2010; Chieffi & Limongi 2002; Limongi & Chieffi 2006, 2012; Nomoto et al. 2006).

4.4.1. SN 1987A. For 13–25- $M_{\odot}$ stars, SN 1987A in the Large Magellanic Cloud has provided the most detailed constraints on the explosion model, although the mechanism to transform the collapse into an explosion remains obscure (Arnett et al. 1989). Nevertheless, we provide some detailed values: (a) Given identification of the blue-supergiant progenitor, the main-sequence mass is estimated to be $\sim 18\text{--}20 M_{\odot}$. (b) Because of the small radius of the progenitor, the light curve of SN 1987A shows details of radioactive decay powers of $^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$ and $^{57}\text{Ni} \rightarrow ^{57}\text{Co} \rightarrow ^{57}\text{Fe}$, which lead to the estimate of the masses of $^{56}\text{Ni} \sim 0.07 M_{\odot}$ and $^{57}\text{Co} \sim 3 \times 10^{-3} M_{\odot}$ [$<^{57}\text{Co}/^{56}\text{Ni}> = (^{57}\text{Co}/^{56}\text{Ni})/(^{57}\text{Fe}/^{56}\text{Fe})_{\odot} = 1.7$].

The mass of ^{58}Ni can also be constrained from the nebular spectrum, and $<^{58}\text{Ni}/^{56}\text{Ni}> \sim 1$ was obtained for SN 1987A. These ratios constrain $Y_{\text{e}} \sim 0.50$ near the mass cut (e.g., Kumagai et al. 1993). For the 20- $M_{\odot}$ model for SN 1987A, these masses correspond to the baryonic mass of 1.61 $M_{\odot}$ at the mass cut (i.e., the neutron star mass) (e.g., Thielemann, Nomoto & Hashimoto 1996). ^{44}Ti is another radioactive nucleus produced in explosive burning. Its mass estimated from the late slow light curve of SN 1987A amounts to $\sim 1 \times 10^{-4} M_{\odot}$. Whether this amount of ^{44}Ti is consistent with the explosion models remains under debate (e.g., Diehl 2013).

4.4.2. SN 1993J and SN 1994I. Similarly, observations and models of nearby supernovae, i.e., Type Ib SN 1993J and Type Ic SN 1994I, show that these core-collapse supernovae have progenitor masses of 13–15 $M_{\odot}$, $E \sim 1 \times 10^{51}$ erg, and $\sim 0.07 M_{\odot}$ ^{56}Ni (e.g., Nomoto et al. 1993, 1994). Explosion models indicate that the mass cut is $M_{\text{r}} \approx 1.3 M_{\odot}$ for 13–15- $M_{\odot}$ stars. Modeling of nearby supernovae suggests that stars in the mass range of 13 $M_{\odot}$ – $M_{\text{NS/BH}}$ undergo neutron star-forming Fe core collapse and induce normal core-collapse supernovae, which produce significant amounts of heavy elements from α -elements to Fe-peak elements.

4.4.3. Yields. Several groups have provided supernova yields tables for this mass range (Rauscher et al. 2002; Kobayashi et al. 2006; Limongi & Chieffi 2006; Nomoto et al. 2006; Tominaga, Umeda & Nomoto 2007; Heger & Woosley 2010; Limongi & Chieffi 2012). **Figure 5** shows abundance patterns ($[X/\text{Fe}] - Z$) of supernova ejecta for a grid of models for $M = 15, 20$, and 25 $M_{\odot}$, $Z = 0$ and 0.02, and $E_{51} = 1$. It also compares the abundance patterns among the models by Heger & Woosley (2002, 2010), Limongi & Chieffi (2006, 2012), Kobayashi et al. (2006), and Tominaga, Umeda & Nomoto (2007), in which ejected mass of ^{56}Ni is set equally. The overall abundance patterns of the yields provided by these groups are approximately consistent, although there are some exceptionally large differences.

In general, a similar metallicity dependence on the odd-even effect is found, although the deficiency of some odd- Z elements, such as N, F, K, Sc, and Mn, needs to be examined separately. We note the following differences among the three groups:

1. The explosions are initiated by a piston (Rauscher et al. 2002, Heger & Woosley 2010), kinetic bomb (Limongi & Chieffi 2006, 2012), and thermal bomb (Kobayashi et al. 2006; Tominaga, Umeda & Nomoto 2007). These different bombs may lead to different entropy productions in the shocked region for the same E (Aufderheide, Baron & Thielemann 1991), thus affecting the abundance of some odd- Z elements, such as K and Sc.

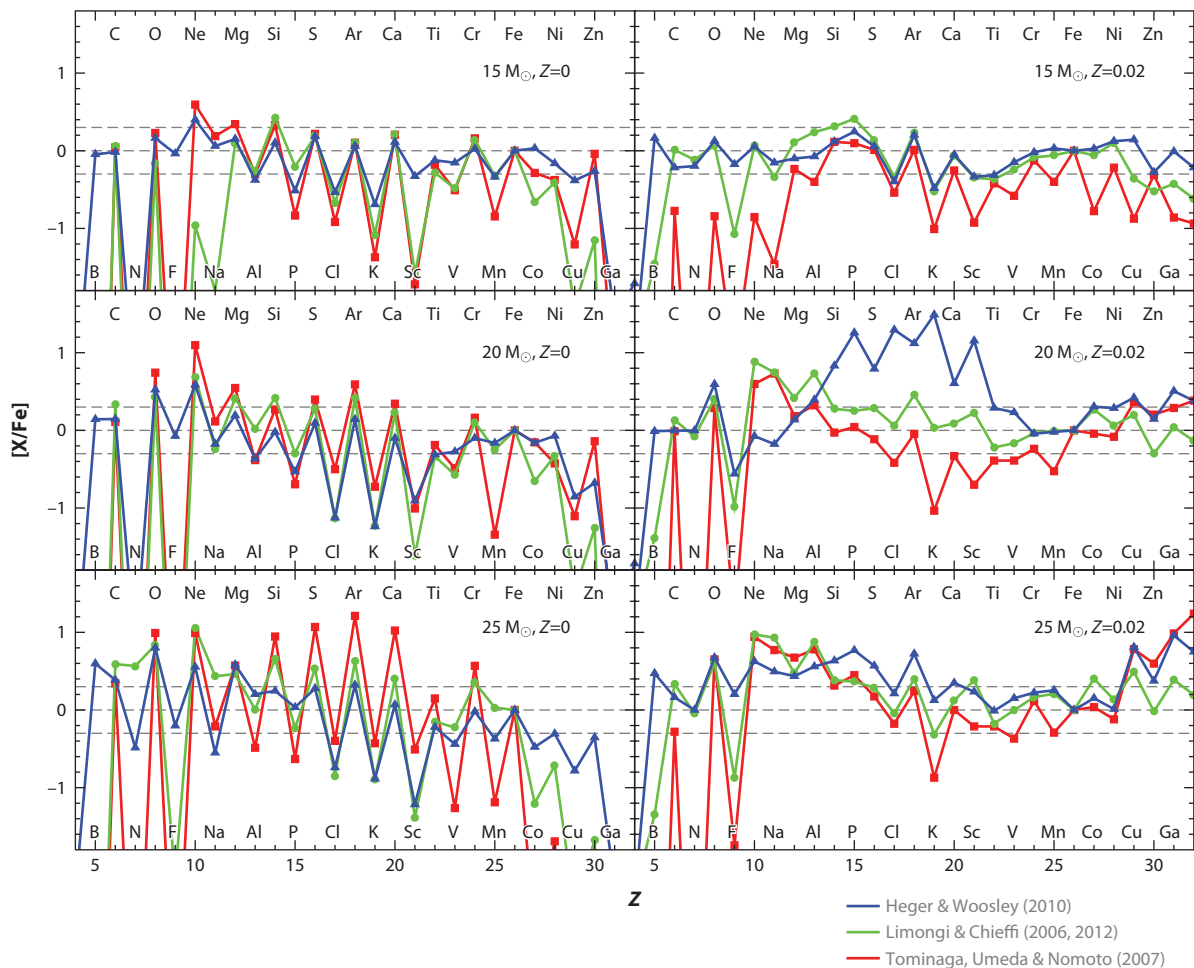


Figure 5

Yields of a grid of models with (*from top to bottom*) $M = 15, 20,$ and $25 M_{\odot}$ and (*from left to right*) $Z = 0$ and 0.02 , respectively, including comparisons among various models: (*blue lines*) Rauscher et al. (2002) and Heger & Woosley (2010); (*green lines*) Limongi & Chieffi (2006, 2012); and (*red lines*) Kobayashi et al. (2006) and Tominaga, Umeda & Nomoto (2007). The overall abundance patterns of yields by these groups are approximately consistent, although there are some exceptionally large differences. The $Z = 0$ and $Z = 0.02$ models show large and small odd-even effects, respectively.

2. The ν process is included in Rauscher et al. (2002), Heger & Woosley (2010), Kobayashi et al. (2011). The ν process enhances the production of F, so that $[F/Fe] \sim 0$ in the models of Rauscher et al. (2002), Heger & Woosley (2010), Kobayashi et al. (2011). The ν process also clearly reduces the odd-even effect for $Z = 0$, although its effect is smaller than the metallicity effect for $Z = 0.02$.

Details of the abundance features of these stellar models are discussed in Sections 7 and 8.

4.5. Hypernovae, Faint Supernovae, and 25–140- $M_{\odot}$ Stars

These stars undergo Fe core collapse to form a black hole. As shown in **Figure 2**, the resulting black hole-forming supernovae seem to bifurcate into two branches: hypernovae and faint supernovae. If

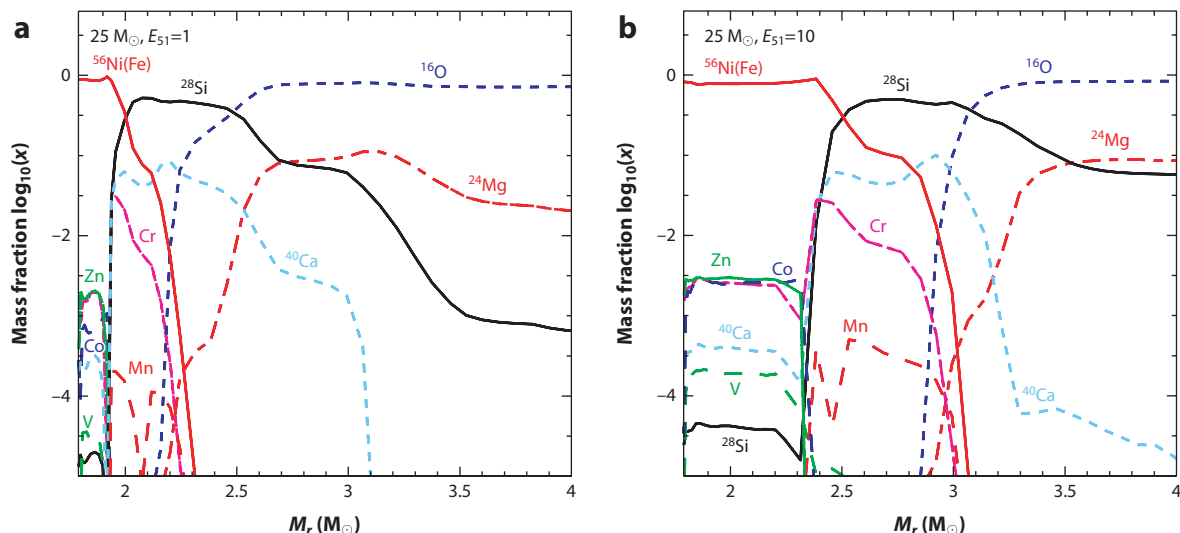


Figure 6

Abundance distribution against the enclosed mass M_r after the explosion of Pop III $25\text{-}M_{\odot}$ stars with (a) $E_{51} = 1$ and (b) $E_{51} = 10$ (Umeda & Nomoto 2002; Tominaga, Umeda & Nomoto 2007): (a) composition of the ejecta for a $25\text{-}M_{\odot}$ hypernova model ($E_{51} = 10$); (b) nucleosynthesis in a normal $25\text{-}M_{\odot}$ supernova model ($E_{51} = 1$) shown for comparison. The high explosion energy in hypernovae causes a shift in the complete and incomplete Si-burning regions outward in mass and enhances α -rich freeze-out in the complete Si-burning layer.

the black hole has little angular momentum, little mass ejection takes place and it would be observed as a faint supernova. By contrast, a rotating black hole could eject matter via jets, resulting in an hypernova. Hypernovae produce a large amount of heavy elements ranging from α -elements to Fe-peak elements.

4.5.1. Hypernovae. We note the following characteristics of nucleosynthesis with very large explosion energies (Nakamura et al. 2001, Nomoto et al. 2001, Umeda & Nomoto 2005) (see **Figure 6**):

1. Compared with normal supernovae, both the complete and incomplete Si-burning regions of hypernovae shift outward in mass, so that the mass ratio between these regions becomes larger. As a result, higher energy explosions tend to produce larger $[(\text{Zn}, \text{Co}, \text{V})/\text{Fe}]$ and smaller $[(\text{Mn}, \text{Cr})/\text{Fe}]$ ratios, which can explain the trend observed in VMP stars (Umeda & Nomoto 2005).
2. In the complete Si-burning region of hypernovae, elements produced by α -rich freeze-out are enhanced. Hence, elements synthesized through the capture of α -particles, such as ^{44}Ti , ^{48}Cr , and ^{64}Ge (decaying into ^{44}Ca , ^{48}Ti , and ^{64}Zn , respectively) are more abundant.
3. O burning takes place in more extended regions for larger E . More O, C, and Al are burned to produce a larger amount of burning products such as Si, S, and Ar. Therefore, hypernova nucleosynthesis is characterized by large abundance ratios of $[\text{Si}, \text{S}/\text{O}]$, which can explain the abundance feature of M82 (Umeda et al. 2002).

4.5.2. Faint supernovae. In contrast to hypernovae, faint supernovae (as shown in **Figure 2**) undergo extensive fallback of processed materials. **Figure 7** shows the abundance patterns of fallback supernovae for various masses and E . Here, the mass cut is set to produce only $\sim 0.01\text{-}M_{\odot}$

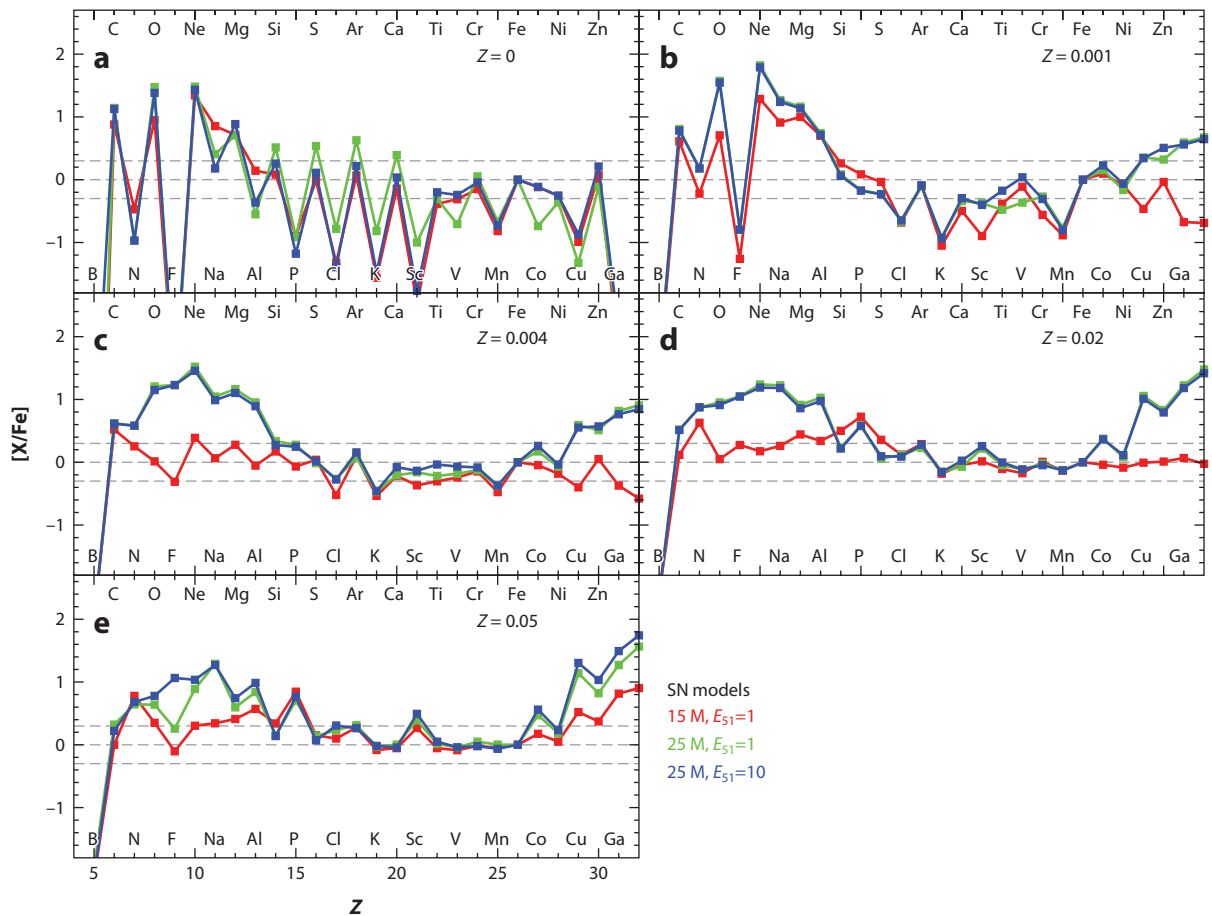


Figure 7

Yields from faint supernova (SN) models for $(M/M_{\odot}, E_{51}) = (\text{red}) (15, 1)$, $(\text{green}) (25, 1)$, and $(\text{blue}) (25, 10)$ with various metallicities of (a) $Z = 0$, (b) $Z = 0.001$, (c) $Z = 0.004$, (d) $Z = 0.02$, and (e) $Z = 0.05$. $[\text{C}/\text{Fe}]$ is higher for lower metallicities because preexisting metals dominate synthesized metals in supernovae with high metallicities.

^{56}Ni , and ejecta of fallback supernovae have large $[\text{C}/\text{Fe}] - [\text{Al}/\text{Fe}]$. These patterns could explain the abundance patterns observed in CEMP stars.

Note that, in spherical explosions, substantial fallback occurs for relatively low E . However, in jet-induced explosions, fallback occurs even for high E . The stellar mass dependence of the abundance pattern is also quite weak. Comparison with observed patterns of CEMP stars makes it difficult to identify the progenitor's mass.

4.5.3. Pulsational instability in precollapse stars. Stars more massive than $M \sim 90 M_{\odot}$ undergo nuclear instabilities and associated pulsations (ϵ -mechanism) at various nuclear burning stages (Heger & Woosley 2002). (a) O-burning: For the $M = 137 M_{\odot}$ Pop III star, the evolutionary track for the central density and temperature is very close to (but outside of) the “ $e^- e^+$ pair-instability region” of $\Gamma < 4/3$, where Γ denotes the adiabatic index (Figure 3) (Ohkubo et al. 2009). During O burning, the central temperature and density of such a massive star oscillate

several times (Woosley, Blinnikov & Heger 2007; Waldman 2008; Ohkubo et al. 2009) because radiation pressure in these stars is so dominant that Γ is close to $4/3$. Thus, the inner core of the stars easily expands in response to the nuclear energy released by O burning. Once the inner core expands, the temperature drops suddenly, the central O burning is weakened, and the stellar core shrinks. Because only a small amount of O is burnt for each cycle, these pulsations occur many times. The amplitude of the temperature and density variations is larger for more massive stars, which suggests increasingly drastic oscillations may occur for larger-mass stars. In extreme cases, pulsation could induce dynamical mass ejection and optical brightening, as may be observed in the brightest SN 2006gy (Woosley, Blinnikov & Heger 2007).

(b) Si-burning: $M \sim 90\text{--}M_{\odot}$ stars also undergo nuclear instability as a result of Si burning, and they pulsate several times (see **Figure 3**) (Umeda & Nomoto 2008). The amplitude of the pulsation due to Si burning in the central density and temperature is smaller than that caused by O burning (Ohkubo et al. 2009). (c) Core-collapse and explosion: Eventually, these $\sim 90\text{--}140\text{--}M_{\odot}$ stars undergo Fe core collapse to form black holes. Hypernova-like energetic supernovae may then occur to produce large amounts of ^{56}Ni . Umeda & Nomoto (2008) found that the synthesized ^{56}Ni mass increases with increasing E and M . For $E = 3 \times 10^{52}$ erg, ^{56}Ni masses of up to 2.2, 2.3, 5.0, and $6.6 M_{\odot}$ can be produced for low-metallicity ($Z = 0.0001$) progenitors with $M = 30, 50, 80$, and $100 M_{\odot}$. Thus, the upper limit to the mass of ^{56}Ni produced by core-collapse supernovae ($M \lesssim 140 M_{\odot}$) would be $\sim 10 M_{\odot}$. The abundance pattern of the ejecta does not depend much on the stellar masses. Because of the large ejecta mass, the expansion velocities may not be high enough to form broad-line features, as has been observed in SN Type Ic 1999as (Nomoto 2012). However, supernovae may become superluminous supernovae when large E and ^{56}Ni mass are present (Moriya et al. 2010a).

5. VERY MASSIVE STARS

5.1. Pair-Instability Supernovae of $140\text{--}300\text{--}M_{\odot}$ Stars

If very massive stars ($M > 140 M_{\odot}$) do not lose much mass, they undergo thermonuclear explosions triggered by pair-creation instability (PISNe) (Barkat, Rakavy & Sack 1967). Such stars are completely disrupted without forming a black hole and, thus, eject a large amount of heavy elements, especially ^{56}Ni (e.g., Umeda & Nomoto 2002, Heger & Woosley 2002). The largest mass of ^{56}Ni obtained in the PISN models amounts to $\sim 40 M_{\odot}$ (Heger & Woosley 2002). The resultant radioactive decays of ^{56}Ni and ^{56}Co could produce superluminous supernovae (Gal-Yam et al. 2009).

Figure 8 shows the abundance patterns in the ejecta of typical models of the core-collapse hypernovae, PISNe, faint supernovae, and Type Ia supernovae. The following PISN abundance features are compared with observed abundances:

- Abundance ratios of Fe-peak elements: $[\text{Zn}/\text{Fe}] < -0.8$ and $[\text{Co}/\text{Fe}] < -0.2$. Such small Zn and Co productions (relative to Fe) are due to the low central temperature at the bounce of the collapsing core (Umeda & Nomoto 2002, Heger & Woosley 2002). This abundance feature of yields is intrinsic to PISNe: If the central temperature becomes higher, photodisintegration effects cause the core collapse to continue.
- Explosive O-burning leads to large $[(\text{Si}, \text{S}, \text{Ca})/\text{O}]$ (~ 0.8).
- The odd-even effect is significantly larger in PISNe than in core-collapse supernovae.

In Yields Table (2013), the yields of PISNe with 170, 200, and $270 M_{\odot}$ have been used (Umeda & Nomoto 2002). The $150 M_{\odot}$ model has been updated on the basis of Umeda & Nomoto (2002). For 140 and $300 M_{\odot}$, the models have been extrapolated.

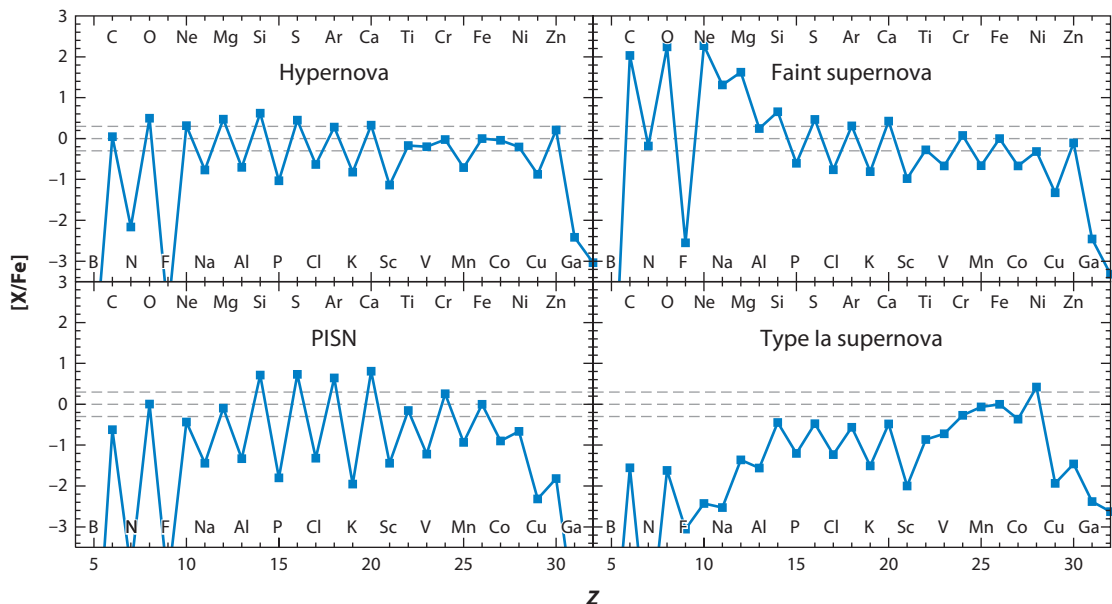


Figure 8

Yields of the core-collapse hypernova, pair-instability supernova (PISN) (Umeda & Nomoto 2002), faint supernova, and Type Ia supernova (Iwamoto et al. 1999) are compared. Faint supernova, PISN, and Type Ia supernova are characterized with higher [C/Fe], lower [Zn/Fe], and larger odd-even effects as well as lower $[\alpha/\text{Fe}]$, respectively, compared with the same ratios of the hypernova.

5.2. Stars with $M \gtrsim 300 M_{\odot}$ and Intermediate-Mass Black Holes

Stars with $300 M_{\odot} < M < 3.5 \times 10^5 M_{\odot}$ enter the pair-instability region but are too massive to be disrupted by PISNe. However, they undergo core collapse, thus forming intermediate-mass black holes (Fryer, Woosley & Heger 2001). If such stars formed rapidly rotating black holes, jet-like mass ejection could occur and produce processed material (Fryer, Woosley & Heger 2001; Ohkubo et al. 2006). Indeed, for moderately aspherical explosions, the patterns of nucleosynthesis show small [O/Fe] and [Ne/Fe] as well as large [Mg/Fe], [Si/Fe], and [S/Fe]. [C/Si] is also not as small as it is in PISNe. These patterns match the observational data of both the intracluster medium and M82 better than do PISNe (Ohkubo et al. 2006).

These results suggest that core-collapse explosions of very massive stars may contribute to chemical enrichment in galaxy clusters. Accordingly, Pop III core-collapse very massive stars may be responsible for the origin of intermediate-mass black holes. Stars more massive than $\sim 3.5 \times 10^5 M_{\odot}$ (supermassive stars) collapse owing to general relativistic instability before reaching the main sequence (e.g., Fowler 1966, Osaki 1966).

6. TYPE IA SUPERNOVAE: PROGENITORS AND NUCLEOSYNTHESIS

The thermonuclear explosion of a C+O white dwarf has successfully explained the basic observed features of Type Ia supernovae. Both the Chandra and the sub-Chandra mass models have been examined (e.g., Livio 2000). However, no clear observational indication reflects how the white dwarf mass grows until C ignition, i.e., whether the white dwarf accretes H/He-rich matter from its binary companion (single-degenerate scenario) or whether two C+O white dwarfs merge (double-degenerate scenario) (e.g., Nomoto 1982a; Iben & Tutukov 1984; Webbink 1984; Arnett 1996;

Nomoto, Iwamoto & Kishimoto 1997; Hillebrandt & Niemeyer 2000; Nomoto et al. 2000, 2009). Several candidate super-Chandra mass explosions have provided important clues to resolve these issues (also see Hachisu et al. 2012 and references therein).

Recent modeling shows that double-degenerate merging could result in both the Chandra and the (effectively) sub-Chandra explosions. The single-degenerate scenario could also result in both explosions. Below, we review potential cross connections between the double- and single-degenerate scenarios and the Chandra and sub-Chandra models and discuss some observational constraints.

1. **Double degenerate and sub-Chandra:** If C detonation is induced near the white dwarf surface in the early dynamical phase, the (effectively) sub-Chandra explosion will result, because the detonated white dwarf has a sub-Chandra mass of $M_1 = 0.9\text{--}1.1 M_\odot$ and a central density as low as $\sim 10^7 \text{ g cm}^{-3}$ (Pakmor et al. 2010).
2. **Double degenerate and Chandra:** If no detonation is induced, the white dwarf could grow until the Chandra mass is reached. The outcome depends on whether the quiescent C shell burning is ignited and burns interior C+O into O+Ne+Mg.
3. **Single degenerate and sub-Chandra:** If the He-shell flashes grow strong enough to induce an He detonation, a sub-Chandra explosion will result (e.g., Nomoto 1982b, Woosley & Weaver 1994).
4. **Single degenerate and Chandra:** If the He-shell flashes are not strong enough to induce an He detonation, they will still produce interesting amounts of intermediate-mass elements, including Si and S, as unburned material near the surface of the C+O white dwarf (Nomoto, Kamiya & Nakasato 2013).

6.1. Chandra versus Sub-Chandra Mass Models

For nucleosynthesis yields, whether the explosion is Chandra or sub-Chandra is crucial, because the central density of the white dwarf affects the abundance ratio of Fe-peak elements. Both Chandra and sub-Chandra explosion models can synthesize relevant amounts of ^{56}Ni for Type Ia supernovae (Hillebrandt & Niemeyer 2000). However, the amount of other Fe-peak elements differs, because the ignition density is different: The density can be as high as $>10^9 \text{ g cm}^{-3}$ in the Chandra model, whereas as low as $\sim 10^7 \text{ g cm}^{-3}$ in the sub-Chandra model.

In the Chandra model, the thermonuclear runaway starts with the ignition of deflagration (e.g., Nomoto, Sugimoto & Neo 1976; Nomoto, Thielemann & Yokoi 1984). In the high-temperature and -density bubble, materials are incinerated into nuclear statistical equilibria and undergo electron capture. Electron capture by free protons and Fe-peak elements leads to the synthesis of ^{58}Ni , ^{54}Fe , and ^{56}Fe (not via ^{56}Ni decay). These neutron-rich Fe-peak elements form a hole that is almost empty of ^{56}Ni (e.g., Nomoto, Thielemann & Yokoi 1984; Höflich et al. 2004).

In the sub-Chandra model, the ignition density is too low for electron capture to take place. The neutron excess is produced only by the initial CNO elements, which are converted to ^{14}N and to ^{22}Ne . Thus, this excess also depends on the initial metallicity. As a result, ^{58}Ni is almost uniformly distributed with a mass fraction as small as ~ 0.01 (e.g., Shigeyama et al. 1992). Such differences in the mass and the distribution of ^{58}Ni can be observationally investigated by late-phase (~ 1 year since the explosion) spectroscopy at near-infrared wavelengths (Höflich et al. 2004). Because ejecta become optically thin at late times, spectroscopy provides an unbiased, direct view of the innermost regions.

Optical observations have also shown $[\text{FeII}] \lambda 7155$ and $[\text{NiII}] \lambda 7378$ for 12 Type Ia supernovae (Maeda et al. 2010b). The $[\text{NiII}] \lambda 7378$ line is emitted from the electron-capture region of the ejecta; the relatively narrow width ($\lesssim 3,000 \text{ km s}^{-1}$) of this line provides further support of this

finding. Thus, the existence of the [NiII] line implies ignition at high density, which supports the Chandra model (Nomoto, Kamiya & Nakasato 2013). In the present chemical evolution model, nucleosynthesis yields of the Chandra mass model W7 have been adopted (Nomoto, Thielemann & Yokoi 1984; Iwamoto et al. 1999). For the single- versus the double-degenerate scenarios, the former has been adopted because of the argument provided in Section 6.2.

6.2. Single- versus Double-Degenerate Scenarios

Some evidence supports the single-degenerate model: for example, the presence of circumstellar matter (Patat et al. 2007, Sternberg et al. 2011, Foley et al. 2012) and the detection of H in circumstellar-interaction type supernovae (Ia/IIn) such as SN 2002ic (Hamuy, Phillips & Suntzeff 2003). In particular, PTF11kx (Dilday et al. 2012) provides strong evidences that the accreting white dwarf was a recurrent nova and the companion star was a red supergiant. However, there has been no direct indication of the presence of companions, e.g., the lack of companion stars in images of SN 2011fe (Li et al. 2011) and some Type Ia supernova remnants (Schaefer & Pagnotta 2012).

In the new single-degenerate models, the accreting white dwarfs are rotating, so that the “effective” Chandra mass is larger than it is in the nonrotating model. Thus, when accretion effectively ceases, the mass of the accreting C+O white dwarf exceeds the Chandra mass to become $1.4\text{--}1.5 M_{\odot}$ in many cases and even larger mass in a few other cases (Hachisu, Kato & Nomoto 2012). Subsequently, the white dwarf undergoes spin-down during which the companion red giant evolves off the red-giant branch to become an He white dwarf. Eventually, the C+O white dwarf explodes after spin-down (Di Stefano, Voss & Claeys 2011; Justham 2011; Hachisu, Kato & Nomoto 2012). The main-sequence companion also loses much mass to become a low-mass main-sequence star or an He white dwarf.

Such He white dwarf companions would be faint enough not to be seen before or after the Type Ia supernova explosion. This new single-degenerate scenario can explain in a unified manner why no signatures of the companion star are seen in some Type Ia supernovae, whereas some Type Ia supernovae indicate the presence of the companion star. In this way, the rotating white dwarf scenario solves the missing-companion problem.

7. CHEMICAL EVOLUTION OF GALAXIES

Different elements are produced from stars on different timescales. Therefore, elemental and isotopic abundance ratios evolve as a function of time in a galaxy. The observed elemental and isotopic abundance ratios can give a constraint on the star formation and chemical enrichment histories of the galaxy. This approach is often called galactic archaeology.

There are various ways of modeling GCE. In so-called one-zone models (also called monolithic models in the cosmological context), it is assumed that the interstellar medium (ISM) of the region under consideration is instantaneously well mixed and has a uniform chemical composition. The formulation of one-zone models has been described in various texts (Tinsley 1980, Pagel 1997, Matteucci 2001) and articles (Prantzos, Casse & Vangioni-Flam 1993; Timmes, Woosley & Weaver 1995; Chiappini, Matteucci & Gratton 1997; Kobayashi, Tsujimoto & Nomoto 2000).

In the following, we use a formula by Kobayashi, Tsujimoto & Nomoto (2000). The parameters are the timescales of star formation and outflow, the fractions of Type Ia supernova progenitors [b_{MS} and b_{RG} in Kobayashi & Nomoto (2009)], IMF (slope, lower limit, and upper limit), and galactic age. These parameters are determined to meet observational constraints such as (a) the gas fraction, (b) the age-metallicity relation (although this is not a strong constraint), (c) the metallicity distribution function (MDF) (the most important constraint), and (d) the [O/Fe]-[Fe/H] relation.

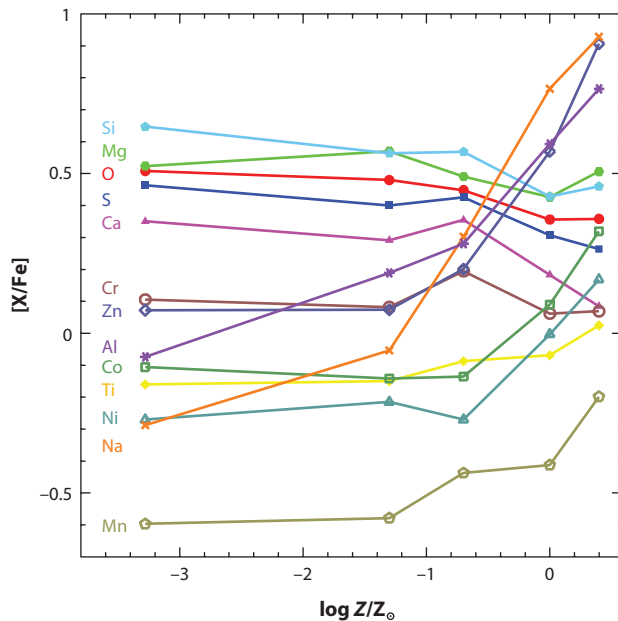


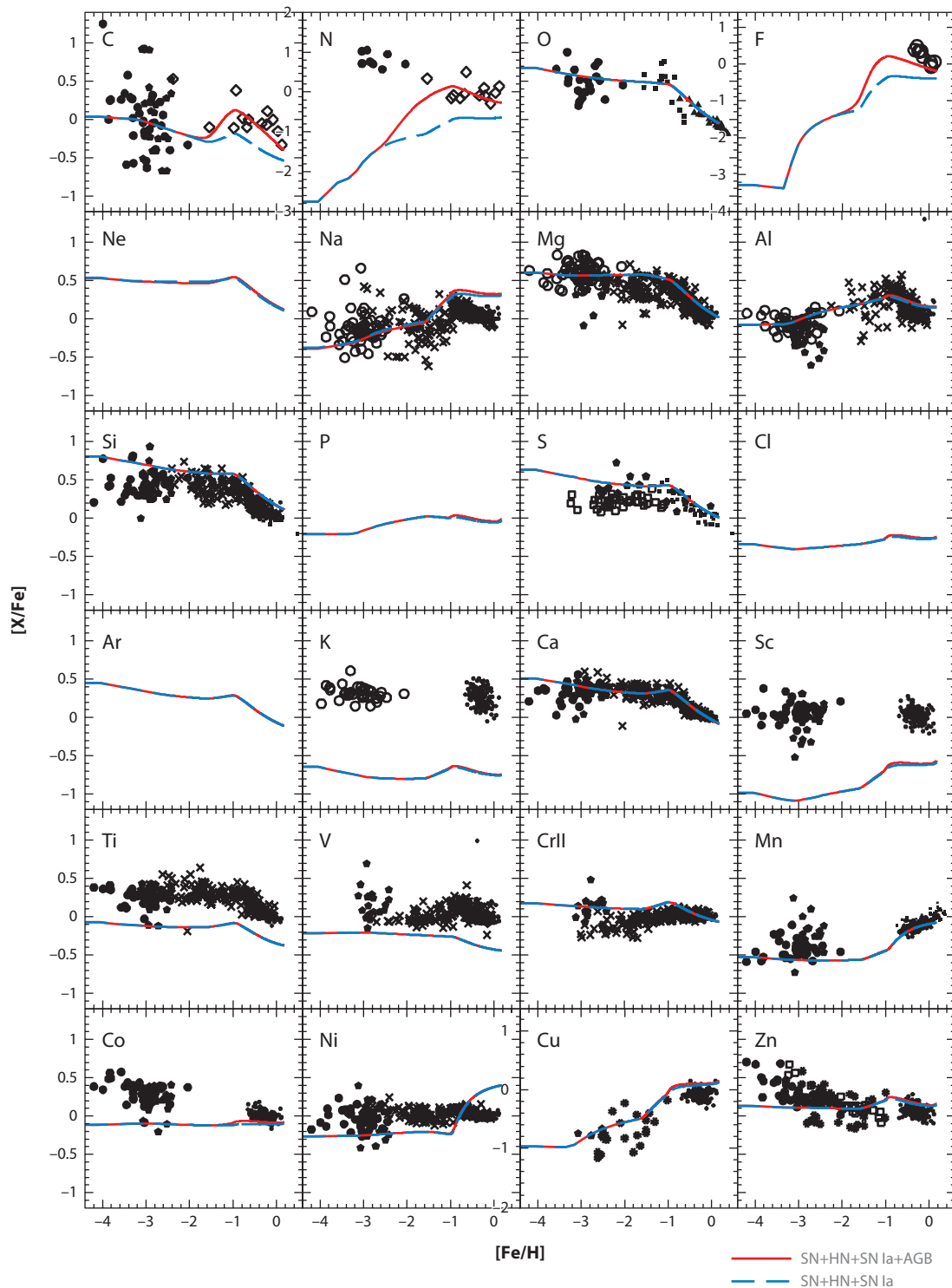
Figure 9

The initial mass function-weighted abundance ratios as a function of the metallicity of the progenitors, wherein the hypernova fraction $\epsilon_{\text{HN}} = 0.5$ is adopted for $M \geq 20 M_{\odot}$. $Z = 0$ results are plotted at $\log Z = -5$. The abundance ratios of odd- Z elements increase at high metallicities owing to a surplus of neutrons, whereas those of even- Z and Fe-peak elements are almost constant over a wide range of metallicities.

Usually, with these observational constraints, the best parameter set can be chosen for a given IMF. The IMF by Kroupa (2008) often gives the best results, but the IMF by Salpeter (1955) can provide almost identical results in terms of elemental abundances. Note that the IMF by Chabrier (2003) causes metal overproduction (for more details, see Kobayashi, Karakas & Umeda 2011).

Figure 9 shows the metallicity dependence of Type II supernova + hypernova yields weighted by the Salpeter IMF at $0.07\text{--}50 M_{\odot}$. In metal-free stellar evolution, because of the lack of initial CNO elements, the CNO cycle does not operate until the star contracts to a much higher central temperature ($\sim 10^8$ K) than that of Pop II stars, at which temperature the $3\text{-}\alpha$ reaction produces a tiny fraction of ^{12}C ($\sim 10^{-10}$ in mass fraction). However, late core evolution and the resulting Fe core masses of metal-free stars are not much different from those of metal-rich stars. Therefore, the $[\alpha/\text{Fe}]$ ratio is larger by only ~ 0.2 dex, and the abundance ratios of the Fe-peak elements are not so different from those of metal-rich stars, except for Mn. On the other hand, the CNO cycle produces only a small amount of ^{14}N , which is transformed into ^{22}Ne during He burning. The surplus of neutrons in ^{22}Ne increases the abundances of odd- Z elements (Na, Al, P, Mn, Cu, etc.). Therefore, the metallicity effect is realized for odd- Z elements, the inverse ratio of α -elements, and their isotopes (e.g., $^{13}\text{C}/^{12}\text{C}$). The $[\text{Na}/\text{Fe}]$ and $[\text{Al}/\text{Fe}]$ ratios of metal-free stars are smaller by ~ 1.0 and 0.7 dex than they are for solar-abundance stars, which is consistent with the observed trends (**Figure 10**).

In reality, the ISM is not well mixed. Thus, more-realistic models have been proposed. Stochastic models (Argast et al. 2002; Ishimaru, Prantzos & Wanajo 2003; Cescutti 2008) can involve inhomogeneous mixing statistically. Hierarchical (semianalytic) models (Tumlinson 2006) include cosmological mass accretion. Recently, it became possible to calculate chemical enrichment



combined with 3D hydrodynamical simulations, which are called chemodynamical simulations (e.g., Kobayashi & Nakasato 2011 and references therein). With chemodynamical simulations, the ages, metallicities, elemental abundances, positions, and kinematics of star particles are obtained. Therefore, the scatter of observational data and the effect of inhomogeneous mixing may be studied. Nevertheless, owing to their short computational times, one-zone models remain useful for quickly checking nucleosynthesis yields. In the following sections, we summarize the GCE of various systems using one-zone models with the Kroupa IMF (for figures, see Kobayashi, Karakas & Umeda 2011; for the Salpeter IMF, see Kobayashi et al. 2006). For most systems (Sections 7.1–7.4 and 7.8), the star-formation rate (SFR) is constrained from the MDF, and for the adopted SFR, elemental abundance ratios are predicted using chemical evolution models. However, in small systems (Sections 7.5–7.7), the SFR is so low that the enrichment source can be derived only with abundance profiling (for relevant background information, see Section 8).

Observations of isotope ratios have opened a new window into studies of the details of stellar evolution, supernovae, and GCE (Kobayashi, Karakas & Umeda 2011). The determination of isotopic ratios from stellar spectra requires very high-quality data, and isotopic determinations are available for only a few elements including Li, C, O, Mg, Si, Ti, Ba, and Eu. Nonetheless, isotope ratios can provide more detailed constraints on the physics of AGB stars and supernovae. Because of differences in their production timescales, isotope ratios can also be used as a tracer of galactic archaeology. In addition, comparing with meteoric data, it may be possible to explore planet formation. For example, Kobayashi, Karakas & Umeda (2011) have shown the time evolution of isotope ratios for the Solar Neighborhood and other Milky Way components (e.g., bulge, thick disk, and halo).

7.1. Solar Neighborhood

GCE in the Solar Neighborhood can be well reproduced with a model that allows infall of material from outside the disk region and star formation taking place over 13 Gyr. Introducing infall significantly reduces the number of metal-poor G-dwarf stars; thus, the MDF shows a narrow distribution peaked at $[\text{Fe}/\text{H}] \sim -0.2$. This is consistent with previous observations (Edvardsson et al. 1993, Wyse & Gilmore 1995), although some recent observations show an even tighter MDF (Holmberg, Nordström & Andersen 2007; Casagrande et al. 2011). The observed age-metallicity relations [Holmberg, Nordström & Andersen 2007; Casagrande et al. 2011; and results from the Radial Velocity Experiment (RAVE) by B. Anguiano & K.C. Freeman, in preparation] show only weak evolution and cannot put a strong constraint on modeling.

Figure 10 shows the evolution of element abundance ratios $[X/\text{Fe}]$ against $[\text{Fe}/\text{H}]$, compared with the observed abundances of stars in the Solar Neighborhood (see also Romano et al. 2010). In both the models and the observational data, the abundances are normalized to the solar abundances of Anders & Grevesse (1989); although Asplund et al. (2009) updated the solar abundances, most

Figure 10

Evolution of elemental abundance ratios $[X/\text{Fe}]$ against $[\text{Fe}/\text{H}]$ for the Solar Neighborhood with (*dashed lines*) only Type II supernovae, hypernovae, and Type Ia supernovae and with (*solid lines*) asymptotic giant branch (AGB) stars. The observational data (*dots*) are taken from the following sources: Cayrel et al. (2004) (*large open circles*); Honda et al. (2004) (*filled pentagons*); Fulbright (2000) (*crosses*); and Reddy et al. (2003), Reddy, Lambert & Prieto (2006), and Reddy & Lambert (2008) for thin (*small filled circles*) and thick (*small open circles*) disk stars. For C and N, only unevolved stars are plotted. For O, see Bensby, Feltzing & Lundström (2004a) for thin (*filled triangles*) and thick (*open triangles*) disk stars. For Si, S, and Zn, see Chen et al. (2002) (*filled squares*), Takada-Hidai et al. (2005) (*filled pentagons*), and Nissen et al. (2007) (*open squares*). Finally, for Cu and Zn, see Primas et al. (2000) (*asterisks*).

observational data are based on one-dimensional (1D) + local thermodynamic equilibrium (LTE) analysis. The 3D corrections of -0.24 for O in C04 and the nonlocal thermodynamic equilibrium (NLTE) corrections of $+0.1$ and $+0.5$ for Mg and Al, respectively, are applied for H04 data (for details, see Kobayashi et al. 2006). For C and N, only data from unevolved stars are plotted. Observational data are taken from several sources that are selected to minimize systematic errors as discussed in Kobayashi & Nakasato (2011). We leave comparison with data of planetary nebulae for future studies. The results are summarized in the following sections.

7.1.1. α Elements. In the early stages of galaxy formation, only Type II supernovae/hypernovae contribute and the $[\alpha/\text{Fe}]$ ratio quickly reaches a plateau ($[\alpha/\text{Fe}] \sim 0.5$). At $[\text{Fe}/\text{H}] \sim -1$, Type Ia supernovae, which produce more Fe than α elements, start to occur. This delayed enrichment of Type Ia supernovae causes the decrease in $[\alpha/\text{Fe}]$ with increasing $[\text{Fe}/\text{H}]$. The $[\text{Fe}/\text{H}]$ where $[\alpha/\text{Fe}]$ starts to decrease depends on the adopted Type Ia supernova progenitor model and is determined not by the lifetime, but by the metallicity dependence of Type Ia supernova progenitors in the model by Kobayashi & Nomoto (2009). As a result, this trend is in excellent agreement with the observations for O, Mg, Si, S, and Ca. Ne and Ar also show a similar trend. As shown in Kobayashi et al. (2006), Ti is underabundant overall, but this problem, as well as that of Sc and V, may be solved with two-dimensional (2D) (Maeda & Nomoto 2003, Tominaga et al. 2007, Tominaga 2009) or 3D (e.g., Janka 2012) nucleosynthesis calculations. AGB stars do not make any difference to these trends. Although AGB stars produce significant amounts of Mg isotopes, inclusion of these does not affect the $[\text{Mg}/\text{Fe}]-[\text{Fe}/\text{H}]$ relation. For O, Israelian, Garcia-López & Rebolo (1998) showed an increasing trend toward lower metallicity with an ultraviolet OH line, but this O trend was due to a 3D effect. When a suitable temperature scale is adopted and NLTE and 3D effects are taken into account, [OI] at $6,300 \text{ \AA}$, OI triplet at $7,774 \text{ \AA}$, and infrared OH lines give consistent results. For S, a similar trend was also found (Israelian & Reboro 2001) and ruled out (for more details, see Kobayashi et al. 2006). For Mg, the abundance measurements of Cayrel et al. (2004) have been updated by Andrievsky et al. (2010), who corrected for the underestimates of equivalent widths. In the model of Kobayashi, Karakas & Umeda (2011), the predicted isotope ratios are roughly consistent with solar ratios, but ^{17}O is overproduced and $^{25,26}\text{Mg}$ are slightly underproduced.

7.1.2. Odd- Z elements. Na, Al, and Cu show a decreasing trend toward lower metallicities; this trend is well reproduced by the strong dependencies of these elements on the metallicities of progenitor stars (Figure 9). In contrast, Na and Al show a decreasing trend toward higher metallicities owing to contributions from Type Ia supernovae; this trend is more shallow than those for α elements. Such a decrease is not seen for Cu, because Cu is also produced by Type Ia supernovae. With updated reactions rates, Na yields from AGB stars are reduced by a factor of ~ 30 (Karakas 2010), and the Na overproduction problem (Fenner et al. 2004) by AGB stars is not seen. Note that the weak s -process of massive stars may produce some Cu (Pignatari et al. 2010), but Cu production from hypernovae should be dominant. K, Sc, and V are underabundant overall, a problem that has not been discussed in detail. The ν -process may increase production of these elements (Kobayashi et al. 2011). $[(\text{P}, \text{Cl})/\text{Fe}]$ are also negative overall in the model predictions. There are metallicity dependencies of P, Cl, K, and Sc yields at $Z > 0.001$ for Type II supernovae/hypernovae, which cause a weak decrease from $[\text{Fe}/\text{H}] \sim -1$ to ~ -3 . V yields depend little on metallicity.

7.1.3. Iron-peak elements. $[(\text{Cr}, \text{Mn}, \text{Co}, \text{Ni}, \text{Zn})/\text{Fe}]$ are consistent with the observed mean values at $-2.5 \lesssim [\text{Fe}/\text{H}] \lesssim -1$. Note that CrII observations are plotted, because this line is not

strongly affected by the NLTE effect (Kobayashi et al. 2006). For Mn, the NLTE effect should not be so large, which is indicated by MnII observations (but see Bergemann & Gehren 2008). As discussed in Kobayashi et al. (2006), compared with the observations, Ni is overproduced by Type Ia supernovae at $[\text{Fe}/\text{H}] \gtrsim -1$, but this problem can be solved by tuning the propagation speed of the burning front and the central density of the white dwarf (Iwamoto et al. 1999) or 3D simulations of thermonuclear explosions (e.g., Röpke et al. 2012). At $[\text{Fe}/\text{H}] \lesssim -2.5$, observational data show an increasing trend of $[(\text{Co}, \text{Zn})/\text{Fe}]$ toward lower metallicities, which are not discussed here because inhomogeneous chemical enrichment is becoming increasingly important. The $[(\text{Co}, \text{Zn})/\text{Fe}]$ trend can be explained by hypernovae under the assumption that observed stars were enriched by only a single supernova (Section 8).

7.1.4. Manganese. Mn is a characteristic element of Type Ia supernova enrichment. Mn production relative to Fe is larger for Type Ia supernovae than for Type Ia supernovae/hypernovae. From $[\text{Fe}/\text{H}] \sim -1$, $[\text{Mn}/\text{Fe}]$ shows an increasing trend toward higher metallicities in observations (Feltzing, Fohlman & Bensby 2007); this trend is caused by the delayed enrichment of Type Ia supernovae (Kobayashi & Nomoto 2009), and not by the metallicity effect that Cescutti et al. (2008) showed with “empirical” yields. In principle, Mn is an odd- Z element, and Mn yields can depend on the metallicities of both Type II supernovae and Type Ia supernovae. However, no strong metallicity dependence is seen in nucleosynthesis yields of Type Ia supernovae (T. Ohkubo, in preparation).

7.1.5. Zinc. Zn is one of the most important elements for supernova physics. $[\text{Zn}/\text{Fe}]$ is ~ 0 for a wide range of metallicities, which can be generated only by a large fraction of hypernovae (50% of $M \geq 20 M_{\odot}$). In detail, there is a small oscillating trend: $[\text{Zn}/\text{Fe}]$ is 0 at $[\text{Fe}/\text{H}] \sim 0$; this ratio increases to 0.2 at $[\text{Fe}/\text{H}] \sim -0.5$, decreases to return to 0 at $[\text{Fe}/\text{H}] \sim -2$, and then increases toward a lower metallicity. This trend is characteristic of the Type Ia supernova model of Kobayashi & Nomoto (2009) and is consistent with observations (e.g., Saito et al. 2009). In theory, Zn production depends on many parameters: ^{64}Zn is synthesized in the deepest regions of hypernovae, whereas neutron-rich isotopes of zinc $^{66-70}\text{Zn}$ are produced by neutron-capture processes, which are larger for massive Type II supernovae with higher metallicities. Compared with solar ratios, ^{64}Zn seems to be underproduced in the models (Kobayashi, Karakas & Umeda 2011). Because the observed $[\text{Zn}/\text{Fe}]$ ratios show an increasing trend toward lower metallicities (Primas et al. 2000, Nissen et al. 2007, Saito et al. 2009), the hypernova fraction may have been larger in the earliest stages of galaxy formation. At higher metallicities, the hypernova fraction may be as small as 1% (Kobayashi & Nakasato 2011).

7.1.6. Carbon. Although the ejected mass of C is similar for low-mass AGB ($\sim 1-4 M_{\odot}$) and massive ($> 10 M_{\odot}$) stars, the $[\text{C}/\text{Fe}]$ ratio is efficiently enhanced by low-mass stars because these stars produce no Fe. The contribution from AGB stars (*solid lines* of **Figure 10**) appears at $[\text{Fe}/\text{H}] \sim -1.5$, which corresponds to the lifetime of $\sim 4 M_{\odot}$ stars (~ 0.1 Gyr) (see also Prantzos, Vangioni-Flam & Chauveau 1994). This behavior is also consistent with the observed behavior of s -process elements (Travaglio et al. 2004). At $[\text{Fe}/\text{H}] \gtrsim -1$, $[\text{C}/\text{Fe}]$ shows a decrease as a result of Type Ia supernovae enrichment. If the yields of rotating massive stars are included, the $[\text{C}/\text{Fe}]$ ratio can be as large as ~ 0.5 at $[\text{Fe}/\text{H}] \sim -2.5$. In observations, a significant fraction of metal-poor stars show a larger C enhancement than ~ 0.5 . Several scenarios, including a single supernova (e.g., Umeda & Nomoto 2002) and AGB stars in binary systems (Suda et al. 2004, Lugaro et al. 2008), have been proposed to explain the observed abundance ratios. Such local peculiar effects are not included in one-zone models. However, AGB stars can contribute at metallicities below $[\text{Fe}/\text{H}] \sim -1.5$ in

the inhomogeneous chemical enrichment of chemodynamical simulations (Kobayashi & Nakasato 2011). Because of the production of ^{13}C from $\sim 4\text{--}7\text{-}M_{\odot}$ AGB stars, the predicted $^{13}\text{C}/^{12}\text{C}$ ratios decrease toward higher metallicities, increase $[\text{Fe}/\text{H}]$ from ~ -1.5 to ~ -0.5 , and then decrease to bring $^{13}\text{C}/^{12}\text{C} = 89$, consistent with the solar ratio (Kobayashi, Karakas & Umeda 2011).

7.1.7. Nitrogen. Different from C, N is produced mainly by intermediate-mass AGB stars ($\sim 4\text{--}7\text{ }M_{\odot}$). Therefore, the AGB contribution (*solid lines* of **Figure 10**) appears at $[\text{Fe}/\text{H}] \sim -2.5$. At $[\text{Fe}/\text{H}] \gtrsim -1$, $[\text{N}/\text{Fe}]$ shows a shallow decrease due to Type Ia supernovae. No difference is seen at $[\text{Fe}/\text{H}] \lesssim -2.5$ for the cases with and without the AGB yields, whereas $[\text{N}/\text{Fe}]$ can be as large as ~ 0.5 with rotating massive stars. Chiappini et al. (2006) have shown that a contribution from rotating massive stars is required to solve the primary N problem. As noted above, however, AGB stars can also contribute to N production even at $[\text{Fe}/\text{H}] \lesssim -2.5$ in chemodynamical simulations (Kobayashi & Nakasato 2011). From the difference between C and N, it is possible to distinguish between contributions from low- and intermediate-mass AGB stars as the enrichment source of the observed metal-poor stars. Given the fraction between C-rich and N-rich stars, the IMF with a Gaussian distribution peaked at $\sim 10\text{ }M_{\odot}$ has been rejected (Pols et al. 2009). The predicted $^{14}\text{N}/^{15}\text{N}$ ratio is too high, probably because of the effects of novae, which likely produced a substantial fraction of the ^{15}N in the Galaxy (Romano & Matteucci 2003).

7.1.8. Fluorine. F is one of the most interesting elements, although F abundances are estimated from only one infrared line in stellar spectra. Both AGB and massive stars have been suggested to produce F, but production has for AGB stars (Jorissen, Smith & Lambert 1992; Abia et al. 2010). The mass range of AGB stars that produce F is similar to the mass range for C and is $\sim 2\text{--}4\text{ }M_{\odot}$. The AGB contribution appears at $[\text{Fe}/\text{H}] \gtrsim -1.5$, but the F production from AGB stars is not enough to explain observations at $[\text{Fe}/\text{H}] \sim 0$ (Cunha et al. 2003). Note that the F yields from AGB stars were increased with the new reaction rates (Karakas 2010). Kobayashi et al. (2011) showed that with the ν -process, F yields from supernovae can be increased by a factor of $\sim 1,000$, which can reproduce the observed F abundances. The ν -process is also expected to be the producer of other elements such as K, Sc, and V. The effect of rotating massive stars is uncertain because F yields are not available in the literature.

7.2. Galactic Bulge

The bulk of stellar populations in the Galactic bulge is old and metal rich, and often considered to be similar to those in elliptical galaxies (e.g., Renzini 1994; Wyse, Gilmore & Franx 1997). McWilliam & Rich (1994) showed a broad MDF that extends to $[\text{Fe}/\text{H}] \sim 1$, which can be reproduced with a simple infall model with a short star-formation timescale. However, Zoccali et al. (2003) showed a narrow MDF with a sharp cutoff at $[\text{Fe}/\text{H}] \sim 0$, which requires a galactic wind possibly driven by supernovae or by feedback from active galactic nuclei. The MDF has been updated by Fulbright, McWilliam & Rich (2006, with high-resolution spectra) and by Zoccali et al. (2008), and the peak metallicity is $[\text{Fe}/\text{H}] \sim 0$, which is higher than the subsolar metallicity in Zoccali et al. (2003), who used a photometric method. There is no young stellar population seen in the color-magnitude diagram (e.g., Zoccali et al. 2003), although a significant young population is found in microlensed dwarf stars (Bensby et al. 2010).

The GCE of the bulge can be well reproduced with an infall + wind model that has a short star-formation timescale (e.g., see model B in Kobayashi et al. 2006). The infall is required to explain the lack of metal-poor stars, and the wind is adopted to quench star formation. It is possible that star formation continues at the present time, forming super-metal-rich stars in the Galactic bulge,

but the SFR should be low to satisfy the MDF. In the infall + wind model, the duration of star formation is set to be 3 Gyr, which results in a maximum metallicity of $[\text{Fe}/\text{H}] \sim +0.3$. A much higher efficiency of chemical enrichment, e.g., a flatter IMF, is not required, unless the duration is much shorter than 3 Gyr. Note that the 3-Gyr duration is consistent with chemodynamical simulations of Milky Way-type galaxies (e.g., Kobayashi & Nakasato 2011).

If the star-formation timescale is shorter than that in the Solar Neighborhood, the contributions from stars of a given lifetime appear at higher metallicities than do the contributions from the stars in the Solar Neighborhood. For example, the plateau of the $[\text{O}/\text{Fe}]-[\text{Fe}/\text{H}]$ relation continues at a supersolar metallicity (see figure 4 in Matteucci & Brocato 1990). At $[\text{Fe}/\text{H}] \gtrsim -1$, $[\alpha/\text{Fe}]$ is higher and $[\text{Mn}/\text{Fe}]$ is lower than they are in the Solar Neighborhood because of the smaller contribution of Type Ia supernovae in the bulge. Simultaneously, $[(\text{C}, \text{N}, \text{F})/\text{Fe}]$ ratios peak at higher metallicities; $[(\text{C}, \text{N}, \text{F})/\text{Fe}]$ is lower at $[\text{Fe}/\text{H}] \lesssim -1$ and slightly higher at $[\text{Fe}/\text{H}] \gtrsim -0.5$ than they are in the Solar Neighborhood. Abundance ratios of $[(\text{Na}, \text{Al}, \text{Cu}, \text{Zn})/\text{Fe}]$ and $[(\text{P}, \text{Cl}, \text{K}, \text{Sc})/\text{Fe}]$ are predicted to be higher because of the metallicity effect on element production from core-collapse supernovae. These model predictions are in good agreement with observations (Fulbright, McWilliam & Rich 2007; Lecureur et al. 2007; Bensby et al. 2009; for a comparison, see Kobayashi & Nakasato 2011) except for the trend associated with $[\text{O}/\text{Mg}]$. The decreasing trend of $[\text{O}/\text{Mg}]$ toward higher metallicity requires some additional effects such as strong stellar winds or a process that causes a change in the C/O ratio (Kobayashi et al. 2006, McWilliam et al. 2008).

The high $[\alpha/\text{Fe}]$ in the bulge can also be realized by changing the IMF, namely, adopting a flatter IMF (see figure 32 of Kobayashi et al. 2006). However, if the bulge stars are more metal-rich than those in the Solar Neighborhood, which has been shown in the updated MDF (Fulbright, McWilliam & Rich 2006; Zoccali et al. 2008), then such a flatter IMF is not required. Johnson et al. (2013) showed that the observed $[(\text{O}, \text{Si}, \text{Ca})/\text{Fe}]$ trends are consistent with the model with the Kroupa IMF. With a flatter IMF, the predicted $[(\text{Zn}, \text{Co})/\text{Fe}]$ ratios become much larger, a result that can be tested in future observations. For neutron-capture elements, unexpected observational results have been reported (Johnson et al. 2012), which may require a variation of chemical enrichment processes.

Elemental abundance ratios are also important to study the origin of the Galactic bulge. In cosmological simulations, assembly of gas-rich small galaxies at $z \gtrsim 2$ results in “classical” bulges (e.g., Kobayashi & Nakasato 2011), whereas secular evolution could produce boxy bulges (e.g., Athanassoula & Misiriotis 2002) with cylindrical rotation (as observed by Howard et al. 2009). Recent observational data suggest two populations in the Galactic bulge: a metal-rich bar and a metal-poor classical bulge (Babusiaux et al. 2010, Hill et al. 2011, Uttenthaler et al. 2012; but see Ness & Freeman 2012). These elemental abundance ratios and their radial and vertical gradients should depend on formation scenarios. The chemodynamical model of Kobayashi & Nakasato (2011) predicted both metallicity and $[\alpha/\text{Fe}]$ vertical gradients in the bulge. These predictions can be tested using Galactic bulge surveys such as the APO Galactic Evolution Experiment (APOGEE).

7.3. Galactic Thick Disk

Galactic thick-disk stars are selected on the basis of their kinematics in the Solar Neighborhood, and formation scenarios have been debated for the following observational features: (a) lack of vertical gradients of metallicity, (b) existence of Type Ia supernova contributions, (c) larger $[\alpha/\text{Fe}]$ than in thin-disk stars, (d) older age than the thin disk, and (e) lack of metal-poor G-dwarfs (e.g., Gilmore, Wyse & Jones 1995; Feltzing, Bensby & Lundström 2003).

The GCE of the thick disk, namely the observed MDF (Wyse & Gilmore 1995) and age-metallicity relation (Bensby, Feltzing & Lundström et al. 2004b), can be well reproduced with an

infall + wind model (e.g., model C in Kobayashi et al. 2006). In this model, the formation timescale is as short as 3 Gyr, and the star-formation efficiency is larger than that of the Solar Neighborhood but smaller than of the bulge. Similar to the bulge model, because the star-formation timescale is shorter and the Type Ia supernova contribution is smaller than in the thin disk, at a given $[\text{Fe}/\text{H}]$, $[\alpha/\text{Fe}]$ is slightly higher and $[\text{Mn}/\text{Fe}]$ is slightly lower than in the thick disk. However, in contrast to the bulge model, the metallicity is not as high, and thus, $[(\text{Na}, \text{Al}, \text{Cu}, \text{Zn})/\text{Fe}]$ are not as high as in the bulge. These are roughly consistent with observations (Prochaska et al. 2000; Bensby, Feltzing & Lundström 2004a; Reddy, Lambert & Prieto 2006; Reddy & Lambert 2008; for a comparison, see Kobayashi & Nakasato 2011). At $[\text{Fe}/\text{H}] \lesssim -1$, because $[\text{X}/\text{Fe}]$ depends only on the IMF-weighted Type II supernova yields in the thick-disk and Solar Neighborhood models, there is no difference in $[\text{X}/\text{Fe}]$ between these two models. This is also consistent with the observation of RAVE-selected metal-poor stars (Ruchti et al. 2010).

Again, elemental abundance ratios are important to constrain formation processes of the thick disk. In cosmological simulations, merging of satellite galaxies leads to the formation of half of the thick-disk stars in the Solar Neighborhood (Kobayashi & Nakasato 2011), whereas radial mixing of stellar populations can also explain the observed $[\text{O}/\text{Fe}]$ relation (Schönrich & Binney 2009). Disk heating indicates a vertical gradient in the rotational velocity (Villalobos & Helmi 2008). Clumpy disks reflect a constant scale height with radius (Bournaud, Elmegreen & Martig 2009). These formation scenarios should predict a difference in radial and vertical gradients in $[\text{X}/\text{Fe}]$, which can be tested with large-scale surveys such as the High-Efficiency and Resolution Multi-Element Spectrograph (HERMES) for the Anglo-Australian Telescope (AAT). The chemodynamical model of Kobayashi & Nakasato (2011) predicted metallicity radial gradients but no $[\alpha/\text{Fe}]$ radial gradients in the disk.

7.4. Galactic Halo

For the Galactic halo, the MDF shows a peak at $[\text{Fe}/\text{H}] \sim -1.6$ (e.g., Chiba & Yoshii 1998; also see the SDSS result by An et al. 2013), which can be reproduced with an outflow model (for the Salpeter IMF, see Kobayashi et al. 2006; for the Kroupé IMF, see Kobayashi, Karakas & Umeda 2011). The star-formation efficiency is much lower than that of the other components, and outflow causes an effective metal loss proportional to the SFR.

In this outflow model, the chemical enrichment timescale of the halo is longer than that of the Solar Neighborhood. Consequently, the $[(\text{C}, \text{F})/\text{Fe}]$ and $^{12}\text{C}/^{13}\text{C}$ ratios are higher, owing to a significant contribution from low-mass AGB stars. Whereas the $[\alpha/\text{Fe}]$ and $[\text{Mn}/\text{Fe}]$ ratios are the same as in the Solar Neighborhood, the $[(\text{Na}, \text{Al}, \text{P}, \text{Cl}, \text{K}, \text{Sc}, \text{Cu}, \text{Zn})/\text{Fe}]$ ratios are predicted to be lower. Using these ratios, it is possible to select stars that formed in a system with a low chemical enrichment efficiency, e.g., the satellite galaxies that were accreted onto the Milky Way Galaxy.

The structures of the Galactic halo may be more complicated. SEGUE observations showed two populations for the Galactic halo and also CEMP fractions that increase at low metallicity; the outer halo has lower metallicity and higher $[\text{C}/\text{Fe}]$ than those in the inner halo (Carollo et al. 2012). These results suggest that faint supernovae were important in the first stages of chemical enrichment (Kobayashi, Tominaga & Nomoto 2011). The Fe-peak and neutron-capture element abundances of these CEMP stars in particular are important for distinguishing this scenario from the other scenario of binary mass transfer.

7.5. Globular Clusters

Globular clusters (GCs) should form in a deep potential well to remove dark-matter components. The present stellar masses and half-light radii are $M_* \sim 10^{4-6} M_\odot$ and 1–35 pc (e.g., Gilmore

et al. 2007), respectively, which imply high densities ($n \sim 1,000 \text{ cm}^{-3}$). Because the contribution from dark matter is small, total mass may be $M_{\text{tot}} \sim 10^{5-9} M_{\odot}$, given the star-formation efficiency in the range of 0.001–0.1. Progenitors of GCs could be more massive if the GCs have lost a large fraction of stars through relaxation, as shown in N-body simulations (Lamers, Baumgardt & Girles 2010). Star formation takes place quickly, which may blow away the ISM and quench star formation. This is consistent with the narrow MDFs and the lack of scatter in the elemental abundance ratios (e.g., Carretta et al. 2009).

However, some elements display variation, notably the O-Na and Mg-Al anticorrelations (e.g., Kraft et al. 1997); there is a primordial population with high O and N along with low Na and Al, and a polluted population with low O and N along with high Na and Al. The polluted stars also include the products of H burning at high temperature ($\sim 6.5 \times 10^7 \text{ K}$), possibly from AGB stars or rotating massive stars (Gratton, Sneden & Carretta 2004). Both AGB and rotating massive stars scenarios seem to require a different process between GCs and field stars; the former scenario needs extra mixing (Ventura & D’Antona 2006), whereas the latter needs a flat IMF (Decressin, Charbonnel & Meynet 2007). Note that AGB yields with an updated reaction rate and without extra mixing are consistent with the observed Na abundances of field stars in the Solar Neighborhood (Kobayashi, Karakas & Umeda 2011). Why are the yields different in GCs?

Additional questions arise if AGB stars contribute to the chemical enrichment of GCs: Why is $[\alpha/\text{Fe}]$ high in the GCs? Why is there no signature of the enrichment from Type Ia supernovae that have comparable lifetimes to those of AGB stars? These questions can be resolved if the Type Ia supernova rate depends on the progenitor metallicity as proposed in Kobayashi et al. (1998).

7.6. Dwarf Spheroidal Galaxies

For dwarf spheroidal galaxies (dSphs), the present stellar masses and half-light radii are $M_* \sim 10^{3-7} M_{\odot}$ and 20–1,000 pc (e.g., Gilmore et al. 2007), respectively, implying much lower densities than found in GCs. Interestingly, the total mass of dSphs is approximately $M_{\text{tot}} \sim 10^7 M_{\odot}$ independent of the stellar mass (Geha et al. 2009). Thus, dark matter dominates in dSphs much more than it does in the Milky Way Galaxy. As shown in observed color-magnitude diagrams (Tolstoy, Hill & Tosi 2009), star formation occurs slowly with a very low rate. Thus, the ISM is likely to be inhomogeneous (e.g., for Carina, see Venn et al. 2012), and the elemental abundance pattern can be used for abundance profiling.

The $[\alpha/\text{Fe}]$ ratios of stars in dSphs such as Fornax and Sagittarius are lower than those of the Galactic halo stars (Tolstoy, Hill & Tosi 2009). This difference is often interpreted to indicate a larger contribution of Type Ia supernovae in dSphs than that of the Solar Neighborhood. However, the observed $[\text{Mn}/\text{Fe}]$ ratios of dSph stars are as low as those of the Galactic halo stars (McWilliam, Rich & Smecker-Hane 2003; Romano, Cescutti & Matteucci 2011), which rejects Type Ia supernova contribution in dSphs. These abundance patterns are more consistent with the yields of low-mass core-collapse supernovae ($\lesssim 20 M_{\odot}$ in Kobayashi et al. 2006). Indeed, the contribution of low-mass supernovae can be dominant in very low SFRs. This possibility of a truncated IMF has been discussed in Tolstoy et al. (2003) and Koch et al. (2008). Also in the Solar Neighborhood, there are some low $[\alpha/\text{Fe}]$ stars, which have kinematics distinct from those of high $[\alpha/\text{Fe}]$ stars and may be debris of accreted satellite. These abundance patterns of Fe-peak elements (low $[(\text{Cu}, \text{Zn})/\text{Fe}]$ and normal $[\text{Mn}/\text{Fe}]$) are also consistent with the yields of low-mass supernovae (Nissen & Schuster 2011, but they made a different interpretation on the abundance patterns).

Recently, a number of dwarf galaxies that are more faint than classical dSphs, called “ultrafaint” dSphs, have been discovered mostly with the SDSS (e.g., Belokurov et al. 2007). Fainter dSphs have lower mean metallicities (Kirby et al. 2008), and at low metallicities, the elemental abundance ratios may become similar to those of halo stars (Frebel, Kirby & Simon 2010) or they may have a large scatter (Aoki et al. 2009). A few EMP stars at $[\text{Fe}/\text{H}] \lesssim -3.5$ in ultrafaint dSphs also show C enhancement (Norris et al. 2010), which may suggest a contribution from faint supernovae.

7.7. Damped Lyman α Systems

Observations of metal-poor damped Lyman α (DLA) systems have opened a new window into studies of the chemical enrichment of the Universe by the first generations of stars. DLAs are quasar absorbers defined by their high column density of neutral H, $\log N(\text{H})/\text{cm}^{-2} \geq 20.3$. They appear to sample a range of galaxy types from extended HI disks to smaller subgalactic-sized haloes, as well as smaller HI clouds within larger galaxies (Wolfe, Gawiser & Prochaska 2005). Large-scale surveys, such as the SDSS, have increased the number of known DLAs. Follow-up high-resolution spectroscopy of the most metal-poor DLAs is of particular interest, because the gas they trace may have been enriched by only a few generations of stars such that the chemical enrichment from the first stars can be studied with abundance profiling. Moreover, measuring elemental abundances in DLAs is straightforward; the only potential complications are line saturation and dust depletion, and both effects are not very important for metallicities $Z \lesssim 1/100 Z_{\odot}$. The chemical evolution of such systems is also relatively simple. By contrast, in more chemically evolved systems, there are uncertainties in the star-formation history, gas inflow and outflow, and large contributions from AGB stars and Type Ia supernovae. As a result, the signatures of the first stars are easily washed out.

Cooke et al. (2010b) reported a very-metal poor DLA with $[\text{Fe}/\text{H}] \simeq -3$, which exhibits strong C enhancement relative to all other available elements, including $[\text{C}/\text{Fe}] \simeq +1.53$. This is reminiscent of the CEMP stars in the Solar Neighborhood. Kobayashi, Tominaga & Nomoto (2011b) showed that the observed abundance pattern of this C-rich DLA is best explained by faint supernovae and not by PISNe. It seems that not all metal-poor DLAs show C enhancement: Another metal-poor DLA has an abundance pattern that is consistent with normal core-collapse supernovae (Cooke, Pettini & Murphy 2012). For further studies, it is important to increase the sample of metal-poor DLAs and to detect heavy elements in metal-poor DLAs.

7.8. Elliptical Galaxies

In elliptical galaxies, elemental abundance ratios can be estimated from integrated absorption lines (Thomas, Maraston & Bender 2003; Graves et al. 2007; Smith et al. 2009). These ratios can be used to constrain the formation processes of elliptical galaxies as well as the variation in the IMF. **Figure 11** shows the model predictions of the $[\text{X}/\text{Fe}]-[\text{Fe}/\text{H}]$ relations for massive (*solid lines*) and less-massive (*dashed lines*) elliptical galaxies. The predictions are obtained with the infall + wind model with the Salpeter IMF, where the star-formation parameters are determined to meet the observed “red and dead” properties of elliptical galaxies. Both models provide a red color ($B-V = 0.95$) at $z = 0$, but the mean stellar metallicity and mean $[\text{O}/\text{Fe}]$ ratio are $\log Z/Z_{\odot} \sim 0$ and -0.2 , $[\text{O}/\text{Fe}] \sim 0.3$ and 0 for massive and less-massive galaxies, respectively.

The predicted abundance ratios are roughly consistent with the observational data (**Figure 11**, *open circles*) (Smith et al. 2009), but the $[\text{C}/\text{Fe}]$ ratios are lower and the $[\text{Mg}/\text{Fe}]$ ratios are higher in the model of massive ellipticals than in the observations, which may suggest a larger contribution from low-mass stars in massive ellipticals. Note that the Ca underabundance problem in ellipticals

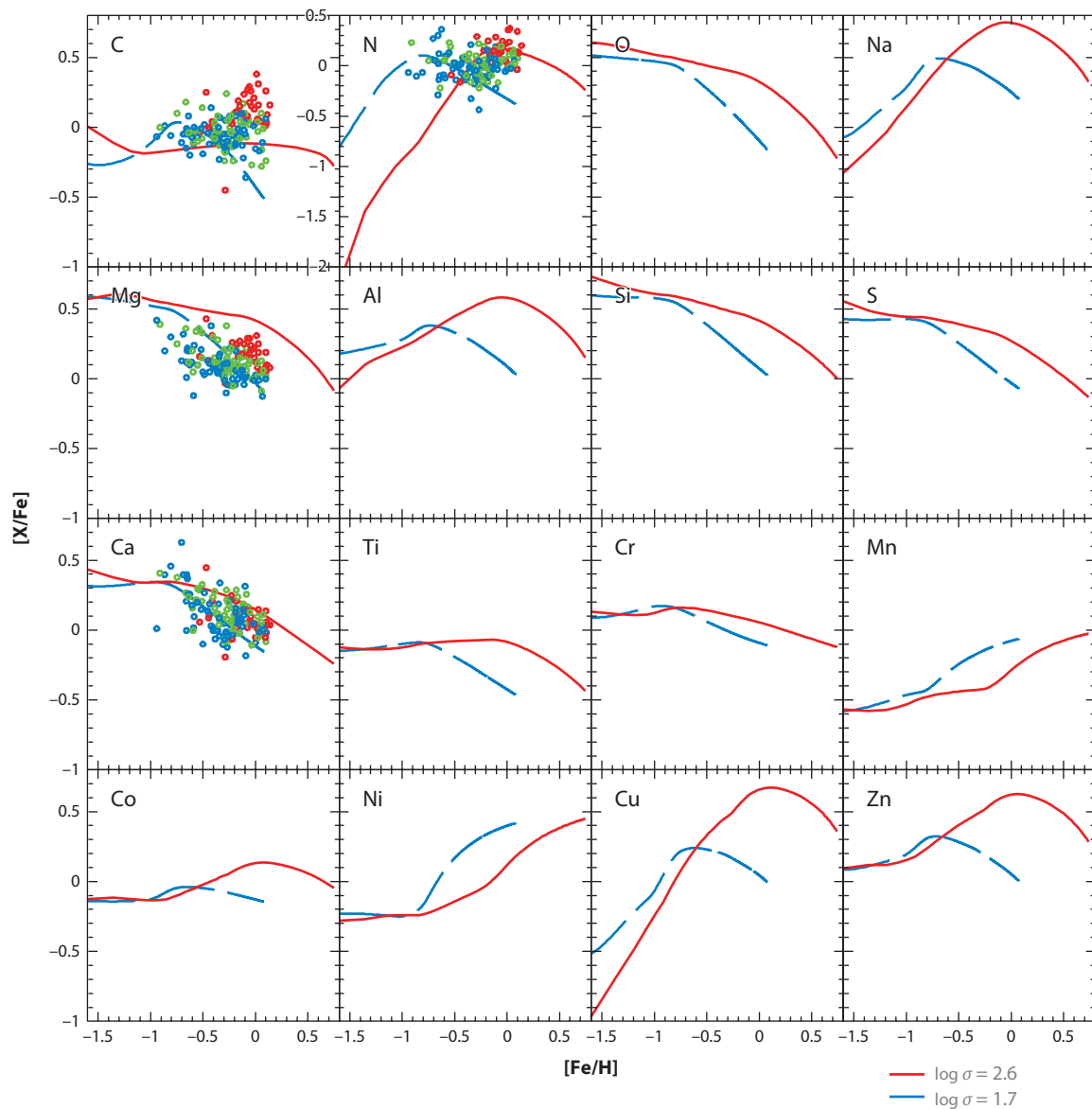


Figure 11

Same ratios as in **Figure 10** but for elliptical galaxies. Solid and dashed lines indicate massive and less-massive galaxies, respectively. Observational data taken from Smith et al. (2009) for $\log \sigma > 2$ (red), > 1.7 (green), and ≤ 1.7 (blue).

(Thomas, Maraston & Bender 2003) was due to the contamination of C and N abundances and is not seen in the analysis method of Smith et al. (2009).

Observationally, chemical compositions have also been estimated for the X-ray hot gas of elliptical galaxies and clusters of galaxies as well as line emissions from active galactic nuclei, and such compositions will be estimated for molecular lines of high-redshift galaxies with ALMA. There

are various uncertainties associated with interpreting these data to derive elemental abundances, and their comparison with theoretical models is left for future studies.

8. EXTREMELY METAL-POOR STARS AND ABUNDANCE PROFILING

Via large-scale surveys and follow-up high-resolution spectroscopy, intensive observations of metal-poor stars have revealed the existence of EMP, UMP, and HMP stars with metallicities as low as $-6 < [\text{Fe}/\text{H}] < -3$ (see Section 1 for definitions). The elemental abundance patterns of these metal-poor stars shed new light on supernova physics such as the mixing and fallback process. Fallback is likely to be the origin of some nearby faint supernovae (e.g., Nomoto et al. 2003) and the C enhancement of CEMP-no stars (see Section 8.3) (e.g., Umeda & Nomoto 2003). However, if all central layers, i.e., the complete Si-burning layers (Section 3.3.1), fall back onto remnants without mixing, the observed elemental abundance ratios in the Solar Neighborhood cannot be explained. Meanwhile, the mixing of elements has been suggested for, e.g., SN 1987A (Kumagai et al. 1988). With mixing (before fallback), the elements synthesized mainly in the complete Si-burning layers, i.e., Zn and Co, can be ejected to match observational data (see Section 7 and **Figure 10**). The mixing and fallback process was first proposed by Umeda & Nomoto (2002) to explain the Zn/Fe ratios in some EMP stars, and it was applied to the stellar yields set in Nomoto et al. (2006) and Kobayashi et al. (2006), whose results are in excellent agreement with the GCE of the Solar Neighborhood. The effect of the mixing and fallback process has also been adopted for the recent $Z = 0$ models by Tominaga, Umeda & Nomoto (2007) and Heger & Woosley (2010).

However, with the one-zone chemical evolution models, it is not possible to discuss the early stages of chemical enrichment where the ISM is likely to be highly inhomogeneous (Section 7). Inhomogeneous enrichment is evidenced by a large scatter of r -process elements (Beers & Christlieb 2005). Furthermore, some stars at $[\text{Fe}/\text{H}] \lesssim -3$ show peculiar abundance patterns that deviate significantly from GCE models. In the early stages of chemical enrichment, stars were enriched predominantly by one or two supernovae (Audouse & Silk 1995, Tumlinson 2006); therefore, the abundance patterns of metal-poor stars can provide important constraints on supernovae nucleosynthesis (abundance profiling). Comparison between the abundance patterns of supernovae models and those of EMP, UMP, and HMP stars can put constraints on nucleosynthesis in Pop III supernovae and, thus, on the nature of Pop III stars.

In inhomogeneous enrichment, metallicity is no longer an indicator of time, but instead reflects only the metallicity of the cloud in which the EMP star formed. The metallicity of the parent cloud of $[\text{Fe}/\text{H}] \lesssim -3$ is dominated by the amount of metal ejected by a supernova (e.g., Audouse & Silk 1995, Tumlinson 2006) and $[\text{Fe}/\text{H}]$ is estimated as follows: The mass of H mixed with supernovae ejecta can be approximated using the mass of H swept up by the supernova shock wave. The swept-up mass is roughly proportional to the explosion energy, E , according to 1D calculations (Thornton et al. 1998, Shigeyama & Tsujimoto 1998). For canonical values of $E = 10^{51}$ erg, $n = 1 \text{ cm}^{-3}$, and ejected Fe mass, $M(\text{Fe}) (= 0.07 M_{\odot})$, the resultant cloud has $[\text{Fe}/\text{H}] \sim -3$, which is consistent with the metallicity of EMP stars. Sophisticated 3D simulations have also been performed (e.g., Nakasato & Shigeyama 2000, Ritter et al. 2012). An average metallicity of a mixture of supernovae ejecta and the pristine gases is $\sim 0.001\text{--}0.01 Z_{\odot}$, which is similar to that obtained via 1D calculations, although there is a wide distribution of $[\text{Fe}/\text{H}]$ ranging from -5 to -1 . These results are highly sensitive to simulation details, such that systematic investigation is required.

Below, we compare observed abundance patterns of C to Zn from stars with $-6 < [\text{Fe}/\text{H}] < -2$ against theoretical stellar yields. For K, Sc, V, and Ti, the reason for the offsets

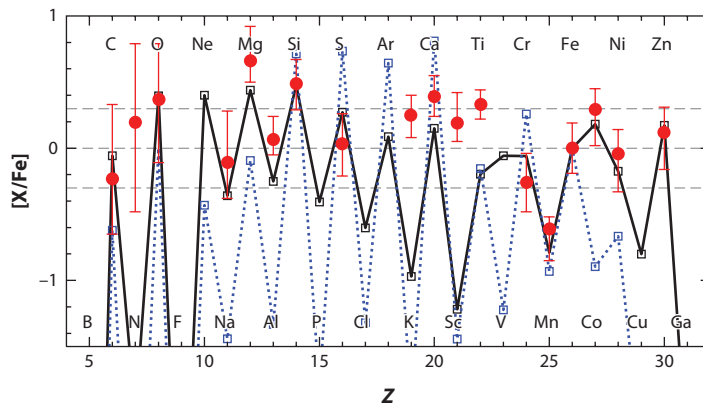


Figure 12

The abundance pattern of very metal-poor stars with $-2.7 < [\text{Fe}/\text{H}] < -2.0$ (red circles with bars) (Cayrel et al. 2004; Andrievsky et al. 2007, 2008; Spite et al. 2011) is in good agreement with the initial mass function integrated yield of Pop III supernovae and hypernovae ranging from $10 M_{\odot}$ to $50 M_{\odot}$ (black solid lines) (Tominaga, Umeda & Nomoto 2007), but not with the $200 M_{\odot}$ pair-instability supernova yield (blue dashed lines) (Umeda & Nomoto 2002).

between nucleosynthesis yields and observations is summarized in Section 7, and approaches to solving these problems are introduced in Section 10.

8.1. Very Metal-Poor Stars

The relatively small scatter of abundance ratios of r -process elements and the metallicity imply that VMP stars with $-3 \leq [\text{Fe}/\text{H}] < -2$ (Beers & Christlieb 2005) are likely to form from gases enriched by many supernovae. VMP stars also likely have an abundance pattern similar to that of well-mixed ejecta of many supernovae. Thus, the one-zone chemical evolution model can be used to explain the abundance ratios (Section 7).

8.1.1. Comparison with normal core-collapse supernova yields. In Figure 12, the abundance patterns of VMP stars [we adopted the averaged abundance pattern of five stars, BD+17:3248, HD 2796, HD 186478, CS 22966-057, and CS 22896-154, all of which have relatively high metallicities ($-2.7 < [\text{Fe}/\text{H}] < -2.0$) (Cayrel et al. 2004)] are compared with supernova and hypernova yields integrated over the progenitors of $M = 10\text{--}50 M_{\odot}$ using the Salpeter IMF (Tominaga, Umeda & Nomoto 2007). The observed and theoretical patterns are in reasonable agreement for many elements, although N, K, and Sc are largely underproduced in the model. For $[\text{N}/\text{Fe}]$, two explanations are possible.

8.1.1.1. Internal origin. As shown in the models, N may be underproduced in Pop III supernovae. However, during the first dredge-up in low-mass red-giant metal-poor stars, N may be enhanced (Suda et al. 2004, Spite et al. 2005, Weiss et al. 2005). Actually, most metal-poor stars are red giants because of their brightness.

8.1.1.2. External origin. N may be enhanced in massive progenitor stars before an supernova explosion. N can be synthesized as a result of mixing between the He convective shell and the H-rich envelope (e.g., Umeda, Nomoto & Nakamura 2000; Iwamoto et al. 2005; Meynet et al.

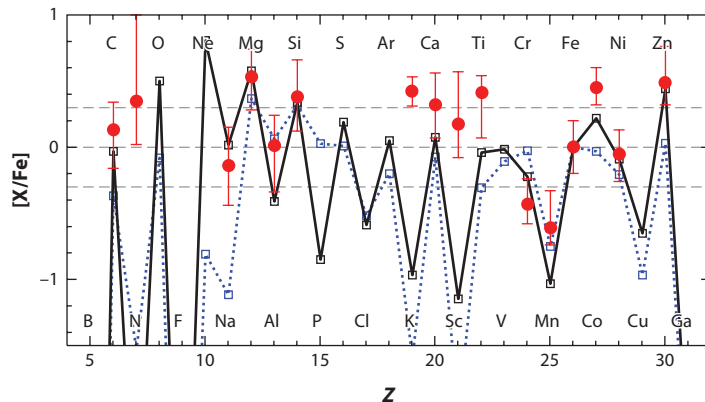


Figure 13

Averaged elemental abundances of stars with $-4.2 < [\text{Fe}/\text{H}] < -3.5$ (red circles with bars) (Cayrel et al. 2004; Andrievsky et al. 2007, 2008, 2010) are compared against normal supernovae (blue dashed lines) ($15 M_{\odot}$, $E_{51} = 1$) and hypernovae yields ($20 M_{\odot}$, $E_{51} = 10$) incorporating the mixing and fallback effects (black solid lines). Hypernovae yields have larger $[(\text{Ti}, \text{Co}, \text{Zn})/\text{Fe}]$ ratios and, thus, are in better agreement with extremely metal-poor stars than with supernovae yields.

2010). Mixing is enhanced by rotation (Langer 1992, Heger & Langer 2000, Maeder & Meynet 2000). If Pop III supernovae progenitors rotate quickly due to small mass loss, then $[\text{N}/\text{Fe}]$ is enhanced, as has been observed in metal-poor stars.

8.1.2. Comparison with pair-instability supernovae yields. Alternatively, VMP stars may form from a cloud enriched by a single supernova if the ejected Fe mass is large enough. In fact, the $200\text{-}M_{\odot}$ PISN produces $7.2 M_{\odot}$ of Fe, which could contaminate a cloud with as much as $[\text{Fe}/\text{H}] \sim -2.5$. In **Figure 12**, the abundance pattern of the yields of the $200\text{-}M_{\odot}$ star is compared with that of VMP stars. The large odd-even effect and low $[\text{Zn}/\text{Fe}]$ of PISNe yields are inconsistent with the abundance pattern of VMP stars. As described in Section 5.1, no verification of the existence of PISNe has been found so far. Further surveys and follow-up spectroscopy are needed to look for a PISN signature.

8.2. Extremely Metal-Poor Stars

As discussed above, EMP stars with $-4 \leq [\text{Fe}/\text{H}] < -3$ are likely to form from the ejecta of a single or a small number of Pop III supernova(e) (e.g., Tumlinson 2006).

8.2.1. Normal supernovae versus hypernovae. **Figure 13** shows a comparison between the averaged abundances of EMP stars ($-4.2 < [\text{Fe}/\text{H}] < -3.5$) [we adopted the averaged abundance pattern of four EMP stars, CS 22189-009, CD-38:245, CS 22172-002, and CS 22885-096, which have low metallicity ($-4.2 < [\text{Fe}/\text{H}] < -3.5$) and normal $[\text{C}/\text{Fe}] \sim 0$ (Cayrel et al. 2004)] and normal Pop III supernovae yields ($15 M_{\odot}$, $E_{51} = 1$) (Tominaga et al. 2007). Supernovae yields are in reasonable agreement with observations for $[(\text{Mg}, \text{Al}, \text{Si})/\text{Fe}]$, but the $[(\text{Mn}, \text{Co}, \text{Zn})/\text{Fe}]$ ratios they give are too small (as described in Timmes, Woosley & Weaver 1995). On the other hand, the larger $[(\text{Ti}, \text{Co}, \text{Zn})/\text{Fe}]$ ratios found in the hypernovae yields ($M = 20 M_{\odot}$, $E_{51} = 10$) in **Figure 13** indicate those yields are in much better agreement with observations than with

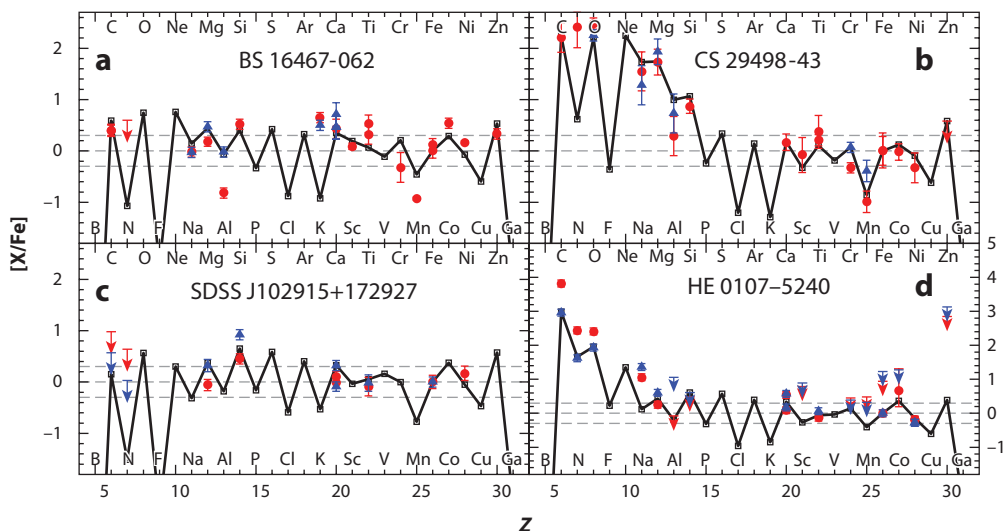


Figure 14

Comparison of observed abundance patterns: 1D-LTE (red filled circles with bars) and 3D-(N)LTE (blue filled triangles with bars), if available, against relevant supernovae models (N. Tominaga, N. Iwamoto, K. Nomoto, submitted). Patterns shown are of the following stars: EMP (BS 16467-062) (Cayrel et al. 2004; Andrievsky et al. 2007, 2008, 2010), CEMP (CS 29498-43) (Aoki et al. 2004), UMP (SDSS J102915+172927) (Caffau et al. 2011, 2012), and HMP (HE 0107-5240) (Bessell, Christlieb & Gustafsson 2004; Collet, Asplund & Trampedach 2006). Supernovae models are constructed as in Tominaga, Umeda & Nomoto (2007) for the $25 M_{\odot}$ progenitor in Iwamoto et al. (2005). Explosion energies and ejected ^{56}Ni masses of supernovae models are $E_{51} = 20$ and $M(^{56}\text{Ni}) = 0.044 M_{\odot}$ (EMP star), $E_{51} = 20$ and $M(^{56}\text{Ni}) = 9.1 \times 10^{-4} M_{\odot}$ (CEMP star), $E_{51} = 20$ and $M(^{56}\text{Ni}) = 1.2 \times 10^{-1} M_{\odot}$ (UMP star), and $E_{51} = 5$ and $M(^{56}\text{Ni}) = 8.0 \times 10^{-5} M_{\odot}$ (HMP star). Abbreviations: 1D, one-dimensional; 3D, three-dimensional; CEMP, carbon-enhanced metal-poor; EMP, extremely metal-poor; HMP, hyper metal-poor; (N)LTE, (non)local thermodynamic equilibrium; UMP, ultra metal-poor.

supernovae yields. Another example of good agreement between an EMP star (BS 16467-062) and an hypernova model is shown in **Figure 14a** (N. Tominaga, N. Iwamoto, K. Nomoto, submitted).

The difference between supernova and hypernova models is seen in the abundance distributions of supernova and hypernova ejecta (**Figure 6**). Both Co and Zn are synthesized in complete Si burning in a high-temperature region, which is more extended in the mass coordinate in the higher E model. **Figure 6** also shows that the mass fractions of Zn, Co, and V in the complete Si-burning region are larger because of higher entropy in the hypernova than in the supernova model. As a result, the integrated ratios of Co/Fe and Zn/Fe are larger in higher-energy explosions (Umeda & Nomoto 2002, 2005; Tominaga, Umeda & Nomoto 2007).

Some have argued that high-energy explosions are not necessary to account for the abundance patterns of EMP stars (Heger & Woosley 2010; Joggerst et al. 2010). However, even when their models with normal explosion energies explain high [Co/Fe], they fail to produce high [Zn/Fe]. These authors proposed that high [Zn/Fe] was due to a contribution from hot bubble or disk wind (see Section 10). However, Izutani & Umeda (2010) showed that hypernovae are necessary to enhance [Co/Fe] and [Zn/Fe] simultaneously without enhancing other Fe-peak elements, as observed in EMP stars (also see **Figure 13**).

8.2.2. Trends with metallicity. Abundance ratios in EMP stars show an interesting metallicity trend, in that $[(\text{Zn}, \text{Co})/\text{Fe}]$ are larger for lower $[\text{Fe}/\text{H}]$ (McWilliam et al. 1995; Ryan, Norris & Beers 1996; Cayrel et al. 2004). In the well-mixed Galaxy model, the metallicity trend is due to the

differences in the lifetimes of progenitors with different masses; however, the model has an issue to resolve, i.e., the scatter of r -process elements (e.g., Cayrel et al. 2004). For the inhomogeneous-mixing model (Argast et al. 2002), Umeda & Nomoto (2005) and Tominaga, Umeda & Nomoto (2007) succeeded in reproducing the observed trends. In their models, hypernovae with higher E and larger Zn/Fe induce the formation of stars with smaller [Fe/H] on average because the ratio of ejected Fe mass to swept-up interstellar H mass is smaller for higher E .

8.2.3. Mixing and fallback. The supernova yields in **Figure 13** include the mixing and fallback effects. In spherical explosion models, fallback depends only on the explosion energy for a given presupernova structure. Thus, lower E explosions lead to more fallback, and no fallback occurs in high E explosions (e.g., Woosley & Weaver 1995; Iwamoto et al. 2005; Fryer, Hungerford & Young 2007).

However, there are problems with the supernovae yields, which cannot simultaneously reproduce $[\alpha/\text{Fe}]$ and $[(\text{Co}, \text{Zn})/\text{Fe}]$ of EMP stars (**Figure 13**). The explosion with $E \lesssim 10^{51}$ erg induces fallback of the inner materials and generates low $[(\text{Co}, \text{Zn})/\text{Fe}]$. By contrast, the explosion with $E \gtrsim 10^{52}$ erg ejects all the materials above the central remnant, enhances $[(\text{Co}, \text{Zn})/\text{Fe}]$ due to explosive nucleosynthesis in the high-entropy environment, and generates low $[\alpha/\text{Fe}]$.

To account for observations, materials synthesized in a deeper complete Si-burning region should be ejected, but the amount of Fe should be small. Ejection of materials with large $[(\text{Co}, \text{Zn})/\text{Fe}]$ (to account for some CEMP stars) also implies that fallback should occur even for high E explosions (see Section 8.3). Hence, Umeda & Nomoto (2002) proposed a phenomenological model called a mixing and fallback model (Section 3.2), which accounts for ejection of the inner matter and fallback of the outer matter and successfully reproduces the abundance patterns of EMP stars (**Figure 13**) (Umeda & Nomoto 2002, 2005; Tominaga, Umeda & Nomoto 2007; Heger & Woosley 2010). This model approximates well the mixing and fallback in the Rayleigh-Taylor instability and in jet-like explosions. In jet-like explosions, fallback occurs even for high E explosions (Tominaga 2009) (see below).

8.3. Carbon-Enhanced Metal-Poor Stars

A significant fraction of metal-poor stars show C enhancement as large as $[\text{C}/\text{Fe}] \geq +1$; hence, they are called CEMP stars (e.g., Beers & Christlieb 2005, Aoki et al. 2008, Cohen et al. 2008, Yong et al. 2013 and references therein). CEMP stars are further classified into CEMP- s stars with enhancement of s -process elements ($[\text{C}/\text{Fe}] > 1$, $[\text{Ba}/\text{Fe}] > 1$, and $[\text{Ba}/\text{Eu}] > +0.5$), CEMP- r stars with enhancement of r -process elements ($[\text{C}/\text{Fe}] > 1$ and $[\text{Eu}/\text{Fe}] > 1$), and CEMP-no stars without enhancement of n -capture elements ($[\text{C}/\text{Fe}] > 1$ and $[\text{Ba}/\text{Fe}] < 0$) (Beers & Christlieb 2005).

8.3.1. Models for carbon enhancement. To determine the origin of such C enhancement, the following suggestions have been made: (a) faint supernovae ejecting small Fe (Umeda & Nomoto 2002, 2005; Tominaga et al. 2007), (b) combined enrichment from a normal supernova and a dark supernova (without Fe ejection) (Limongi, Chieffi & Bonifacio 2003), (c) mass loss from a rotating star (Meynet, Ekström & Maeder 2006), (d) transfer of C-rich materials from an AGB binary companion (e.g., Suda et al. 2004, Suda & Fujimoto 2010), and (e) self-enrichment in a metal-poor red-giant star (Fujimoto, Ikeda & Iben 2000; Campbell, Lugaro & Karakas 2010).

8.3.1.1. Faint supernovae. Faint supernovae occur as a result of fallback of a large amount of radioactive ^{56}Ni whose decay into Fe powers the supernova light curve (**Figure 2**). The fallback

of C is less than that of Fe because of the outer location of C. Accordingly, large $[C/Fe]$ results (Umeda & Nomoto 2003, 2005). **Figure 14b** shows that the abundance pattern of a CEMP star (CS 29498-043) (Aoki et al. 2004) is well reproduced by a single faint supernova ($M = 25 M_{\odot}$) (N. Tominaga, N. Iwamoto, K. Nomoto, submitted). The physical mechanism of mixing and fallback that produces $[C/Fe] > 1$ is a jet-like explosion (Tominaga et al. 2007) rather than a Rayleigh-Taylor instability resulting in $[C/Fe] \lesssim +0.5$ (Joggerst et al. 2010). Most CEMP stars show $[O/Mg] > 1$. Faint supernovae enhance $[O/Fe]$ more than $[Mg/Fe]$ because Mg is synthesized in the inner region. As a result, more Mg than O falls back onto the central remnant. Note that the abundance determination of O is subject to uncertain 3D effects (Nissen et al. 2002).

8.3.1.2. Dark and normal supernovae. The model of Limongi, Chieffi & Bonifacio (2003) proposes concurrent enrichment of a cloud by at least two supernovae: One supernova is normal and ejects Fe-peak elements, whereas the other is dark with strong fallback and ejects C without Fe. The star formed from such a cloud would look like a CEMP star.

8.3.1.3. Mass loss from a rotating star. The model of Meynet, Ekström & Maeder (2006) proposes that a rapidly rotating star with $Z = 10^{-8}$ experiences strong internal mixing and convective dredge-up to enrich the stellar surface with CNO elements. A large amount of mass loss from such a star can enhance C in a cloud. If this cloud is also enriched by a normal (or faint) supernova, the metal-poor star formed from that cloud could look like a CEMP star. However, mass loss of rapidly rotating $Z = 0$ stars is small, whereas rotational mixing enhances ^{14}N in the envelope (Ekström et al. 2008).

8.3.1.4. Mass transfer from an AGB binary companion. Mass transfer from a thermal-pulsing AGB star enhances C and s-process elements simultaneously in the surface abundance of a metal-poor star and transforms a metal-poor star to a CEMP-s star (e.g., Iwamoto et al. 2004, Suda et al. 2004). A signature of binarity may be seen statistically for all CEMP-s stars (Lucatello et al. 2005). This process is also suggested as the origin of CEMP-no stars, because enhancement of s-process elements may be suppressed by an inefficient n supply (Komiya et al. 2007). Whether this process can explain CEMP-no stars is under investigation (Suda et al. 2004, Suda & Fujimoto 2010, Lugaro et al. 2012).

8.3.1.5. Self-enrichment. He-core flash in a metal-poor red giant causes a mixing of H and core material and can enhance the CNO elements on the surface of a red giant (Fujimoto, Ikeda & Iben 2000; Campbell, Lugaro & Karakas 2010). Some n -capture elements can be produced with a supply of free neutrons via $^{13}\text{C}(\alpha, n)^{16}\text{O}$ in the core and mixed to the surface. The resultant enrichment may explain the abundance features of CEMP stars (Campbell, Lugaro & Karakas 2010).

8.3.2. Comparison of carbon-enhancement models. The various models provide different accounts of the origin of elements other than CNO. The faint-supernovae model attributes the origin of CEMP stars to a single Pop III supernova, whereas the other models explain C enhancement but require (an)other supernova(e) to produce elements heavier than CNO and/or Fe-peak elements. The process associated with Fe-peak elements may be a concurrent pollution by a normal supernova or an hypernova (Limongi, Chieffi & Bonifacio et al. 2003) and/or mass accretion from the ISM (Yoshii 1981). Mass accretion from the ISM could also produce a CEMP star if the ISM is C rich.

Only the model of mass transfer from an AGB companion accounts for the origin of CEMP-*s* stars. However, the lowest $[\text{Fe}/\text{H}]$ of CEMP-*s* stars is -3.1 (Johnson & Bolte 2002). The origin of CEMP-no stars, which compose the dominant population at low metallicity, and of CEMP-*r* stars remains under debate. However, a CEMP-no star providing clues has been found: BD+44° 493 shows high $[\text{C}/\text{N}]$ (>0), low $[\text{C}/\text{O}]$ (<0), low $[\text{Ba}/\text{Fe}]$ (<0), and a low upper limit on $[\text{Pb}/\text{Fe}]$, which can be reproduced only by the faint-supernovae model (Ito et al. 2009, 2013). This indicates that at least some CEMP-no stars originate from gas enriched by a faint supernova. Future observations could place further constraints on the origin of CEMP-no stars.

8.4. Nitrogen-Enhanced Metal-Poor Stars

Nitrogen-enhanced metal-poor (NEMP) stars show an enhancement of N, with $[\text{N}/\text{Fe}] > +0.5$ and $[\text{C}/\text{N}] < -0.5$ (e.g., Spite et al. 2005, 2006; Johnson et al. 2007). The most metal-poor NEMP star is CD-38:245 ($[\text{Fe}/\text{H}] = -4.19$, Cayrel et al. 2004). N enhancement may be realized via the following mechanisms:

1. Mixing enhances conversion from C to N inside NEMP stars and may be the dominant process producing NEMP stars at $[\text{Fe}/\text{H}] \lesssim -3$ (Spite et al. 2005, 2006). Indeed, half of NEMP stars display a signature of mixing in Li abundance and $^{13}\text{C}/^{12}\text{C}$ (Spite et al. 2005, 2006).
2. If mixing in the NEMP star were inefficient, mixing in the AGB companion star could enhance the ratio up to $[\text{N}/\text{C}] \sim +2$ in its envelope. Via mass transfer, the AGB companion star could then enhance $[\text{N}/\text{C}]$ in the NEMP star, as is the case for CEMP-*s* stars (e.g., Nishimura et al. 2009). Mixing in the parent massive star enhances N only up to $[\text{N}/\text{C}] \sim +0.5$; thus, it cannot explain NEMP stars (Iwamoto et al. 2005; Meynet, Ekström & Maeder 2006).

8.5. Ultra Metal-Poor and Hyper Metal-Poor Stars

Despite surveys of intensive metal-poor stars, only five stars have been discovered at $[\text{Fe}/\text{H}] < -4$: CD-38:245 ($[\text{Fe}/\text{H}] = -4.19$) (Cayrel et al. 2004), HE 0557-4840 ($[\text{Fe}/\text{H}] = -4.75$) (Norris et al. 2007), SDSS J102915+172927 ($[\text{Fe}/\text{H}] = -4.89$) (Caffau et al. 2011), HE 0107-5240 ($[\text{Fe}/\text{H}] = -5.2$) (Christlieb et al. 2002), and HE 1327-2326 ($[\text{Fe}/\text{H}] = -5.4$) (Frebel et al. 2005). It remains to be established whether the paucity of stars is due to the cutoff in the MDF or to the low-number tail of the MDF at low metallicity (e.g., Yong et al. 2013). The MDF at low metallicity is an interesting issue to be investigated in future surveys. In the following sections, we focus on the abundance patterns of two UMP stars and two HMP stars with $[\text{Fe}/\text{H}] < -4.5$. Observed abundance patterns of UMP (Caffau et al. 2011) and HMP stars (Christlieb et al. 2004) compared with theoretical models are shown in **Figure 14c,d** (for details regarding supernovae models, see N. Tominaga, N. Iwamoto, K. Nomoto, submitted).

8.5.1. Hyper metal-poor stars. The discoveries of two HMP stars have raised an important issue regarding whether the observed low-mass ($\sim 0.8\text{-}M_{\odot}$) HMP stars are Pop III stars or the second-generation stars formed from gases that were chemically enriched by a single Pop III supernova (Umeda & Nomoto 2003). This issue is related to a question of how the IMF depends on metallicity. Thus, identifying the origin of these HMP stars is indispensable for understanding the earliest star formations and the chemical enrichment history of the Universe.

Elemental abundance patterns of these HMP stars provide a key to answering to the above questions. The abundance patterns of HE 1327-2326 (Frebel et al. 2005) and HE 0107-5240

(Bessell, Christlieb & Gustafsson 2004; Christlieb et al. 2004; Collet, Asplund & Trampedach 2006) are quite unusual. The striking similarity of $[\text{Fe}/\text{H}]$ (-5.4 for HE 1327–2326 and -5.2 for HE 0107–5240) and $[\text{C}/\text{Fe}]$ ($\sim +4$) suggests that similar chemical enrichment mechanisms operated during the formation of these HMP stars. However, the N/C and $(\text{Na}, \text{Mg}, \text{Al})/\text{Fe}$ ratios are more than a factor of 10 larger in HE 1327–2326. For the theoretical models to be viable, these similarities and differences must be explained self-consistently. The five models for CEMP stars described above also provide the mechanisms for reproducing the peculiar abundance patterns of HMP stars.

8.5.1.1. Faint supernovae. Iwamoto et al. (2005) showed that the above-mentioned similarities and variations in HMP stars can be well reproduced in a unified manner via nucleosynthesis in core-collapse “faint” supernovae, which undergo mixing and fallback (Umeda & Nomoto 2003). Associated abundance patterns have been also reproduced using jet-induced explosions (Tominaga et al. 2007). Thus, Iwamoto et al. (2005) have argued that HMP stars are the second-generation low-mass stars, whose formation was induced by a Pop III supernova with efficient cooling of C-enriched gases.

8.5.1.2. Rotational mass-loss or dark-supernova models. Both mass-loss and dark-supernova models predict low $[\text{C}/\text{O}]$ (~ -0.2) (Limongi, Chieffi & Bonifacio 2003; Meynet, Ekström & Maeder 2006), which is consistent with HE 1327–2326 but not with HE 0107–5240. Although Limongi, Chieffi & Bonifacio (2003) showed low $[\text{N}/\text{C}]$ (~ -1.5), N abundance can be enhanced if the progenitor star was rapidly rotating (Iwamoto et al. 2005).

8.5.1.3. Mass transfer from a binary companion. Suda et al. (2004), Nishimura et al. (2009), and Suda & Fujimoto (2010) have proposed that the abundance patterns of HE 0107–5240 and HE 1327–2326 are explained by mass transfer from AGB companions with different masses. They also suggested that Ba was not enhanced due to inefficient s -processes or synthesis of heavy s -process elements such as Pb. However, no signature of the binary companion has been found so far, although the constraint on the binary separation is consistent with model predictions (Suda et al. 2004). HMP stars could be Pop III stars if C is enhanced by this mechanism and the stars accrete metals from the ISM enriched by other supernovae.

8.5.1.4. Self-enrichment. Campbell, Lugaro & Karakas (2010) have suggested a method for reproducing the CNO overabundance and Sr abundance of HE 1327–2326: Stars with an initial metallicity of $[\text{Fe}/\text{H}] = -6.5$ may self-enrich CNO and Sr and reproduce the abundance pattern of HE 1327–2326 with a heavy dilution, e.g., due to mass transfer from the binary companion. The authors did not discuss HE 0107–5240.

8.5.2. Ultra metal-poor stars. In contrast to HMP stars, abundance patterns of UMP stars are similar to those of EMP or CEMP stars, except for $[\text{Fe}/\text{H}]$. Thus, the formation scenarios for EMP or CEMP stars are applicable to UMP stars (Figure 14c). Three-dimensional calculations of the expansion of supernova ejecta into cloudy ISM have shown that the resultant metallicity of the clouds takes a wide range of $[\text{Fe}/\text{H}]$ from -5 to -1 (Ritter et al. 2012). Accordingly, the low $[\text{Fe}/\text{H}]$ cloud may form UMP stars. The same supernova is involved in the enrichment of UMP as well as EMP and CEMP stars.

8.6. Yields from Aspherical Explosions and Peculiar Extremely Metal-Poor Stars

The success of the mixing and fallback model indicates that an aspherical explosion must be considered to explain the abundance patterns of EMP stars. Indeed, polarization and nebular spectral shapes indicate aspherical explosions for nearby core-collapse supernovae (Leonard et al. 2006, Maeda et al. 2008, Wang & Wheeler 2008, Tanaka et al. 2012). Calculations of nucleosynthesis in aspherical explosions have been performed for bipolar (jet-like) explosions (Nagataki 2000, Maeda et al. 2002, Maeda & Nomoto 2003, Tominaga et al. 2007, Tominaga 2009) and for the Rayleigh-Taylor instability (e.g., Kifonidis et al. 2006, Joggerst et al. 2010).

Yields of aspherical explosions well reproduce the abundance patterns of EMP stars (Maeda & Nomoto 2003, Tominaga et al. 2007, Tominaga 2009, Joggerst et al. 2010). Jet-like explosions explain the large variations of $[C/Fe]$ for EMP, CEMP, and HMP stars as being a result of the variation of the energy deposition rate, i.e., jet strength (Tominaga et al. 2007). Good agreement here provides physical support for the mixing and fallback model. EMP and CEMP/HMP stars correspond to models with strong and weak jets, respectively. Jet strength may depend on the neutrino luminosity in the neutrino-annihilation-driven explosion (e.g., MacFadyen, Woosley & Heger 2001) and the stellar rotation and magnetic field strength in the magneto-driven explosion (e.g., Takiwaki, Kotake & Sato 2009). By contrast, C enhancement ($[C/Fe] > +1$) is difficult to reproduce using the Rayleigh-Taylor instability (Joggerst et al. 2010).

Tominaga (2009) has compared the yield of the 2D jet-induced explosion model with that of the 1D mixing and fallback model and demonstrated that the mixing and fallback model captures well the abundance features of the jet-induced explosion. However, compared with the spherical model, the 2D jet-like explosion model produces a higher entropy region because of the energy concentration along the jet axis; this difference holds even for the same explosion energy. Thus, compared with the 2D jet-like explosion model, the 1D mixing and fallback model tends to underproduce entropy-sensitive elements, such as Sc, Ti, Cr, Co, and Zn, that are synthesized in the Si-burning layer (Tominaga 2009). Compared with the 1D model, enhancement of $[(Sc, Ti, Cr, Co, Zn)/Fe]$ in the 2D jet-like explosion model improves agreement with EMP stars. Such enhancement could be important for solving the problem related to the production of ^{44}Ti (Diehl 2012). By contrast, the Rayleigh-Taylor instability in normal supernovae may be well reproduced by the mixing and fallback model because entropy should be similar to that of the spherical model.

Another difference between jet-like explosions and Rayleigh-Taylor instabilities is the abundance distribution of supernova ejecta. In jet-like explosions, the shock wave is stronger along the jet axis and heats up the stellar material to temperatures higher than those found along the equatorial direction. Temperatures are lower along the equatorial plane because shocks are weaker, and densities are higher owing to mass accretion. Therefore, ^{56}Ni and unburned materials (mainly O) are located along the jet axis and the equatorial plane, respectively. By contrast, the Rayleigh-Taylor instability has a smaller angular dependence.

The abundance distribution in a jet-like explosion may explain why stars show the peculiar abundance pattern, e.g., the Si-deficient metal-poor star HE 1424-0241 (Cohen et al. 2007). HE 1424-0241 shows an abundance pattern with high $[Mg/Si]$ (~ 1.4) and normal $[Mg/Fe]$ (~ 0.4), which is difficult to reproduce with previous supernovae models (e.g., Woosley & Weaver 1995, Umeda & Nomoto 2005). Tominaga (2009) suggested that an angle-delimited yield consisting of Mg in the inner region and Fe in the outer region may explain the high $[Mg/Si]$ and normal $[Mg/Fe]$ ratios (**Figure 15**). Such aspherical explosions may produce interesting remnants. For example, morphology and the abundance ratios of (Si, S, Ar, Ca)/Fe suggest that jet-induced explosions may produce the supernova remnant W49B (Lopez et al. 2013). Further studies will

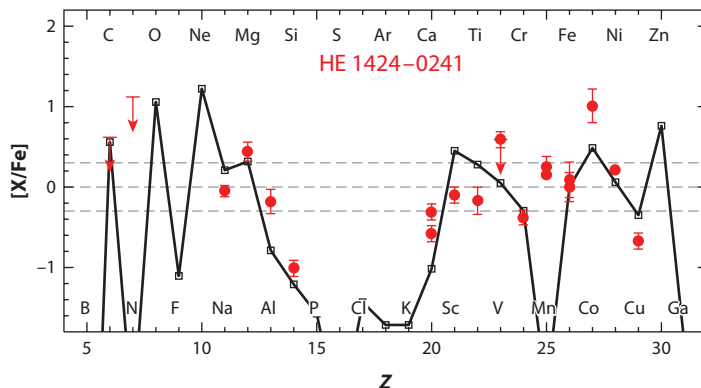


Figure 15

Comparison between the abundance pattern of HE 1424–0241 (Cohen et al. 2007) (*red filled circles*) and angle-delimited yields (*solid line*) (Tominaga 2009). Yields are obtained by integrating the ejecta of jet-like explosions over $30^\circ \leq \theta < 35^\circ$. The abundance pattern of HE 1424–0241 is successfully explained by taking into account the angular dependence of the yields.

be important to determine how the aspherical elemental distribution is maintained during 3D evolution of an supernova remnant in inhomogeneous ISM.

Measurements of detailed abundance patterns have revealed additional EMP stars. These findings have facilitated the discovery of EMP stars with peculiar abundance patterns, e.g., the Si-deficient star HE 1424–0241 (Cohen et al. 2007) and the Ca-rich star SDSS J234723.64+010833.4 (Lai et al. 2009). Although angle-dependent yields explain the abundance pattern of HE 1424–0241, whether angle dependency always works well remains unclear. Thus, the origin of individual EMP stars must be investigated in more depth. A comparison between these peculiar abundance patterns and theoretical supernova models may provide a clue to understanding the diversity of supernovae and chemical evolution in the early Universe.

9. GALACTIC ARCHAEOLOGY SURVEYS

To untangle the star-formation and chemical enrichment histories of the Milky Way Galaxy, a homogeneous and nonbiased sample of kinematics and elemental abundances of millions of stars are needed. Space astrometry missions (e.g., GAIA), large-scale low-resolution spectral surveys (SEGUE and RAVE), and high-resolution spectral surveys will produce unprecedented information regarding the chemodynamical structure of the Milky Way Galaxy (for details, see Ivezić, Beers & Jurić 2012). Future projects involving high-resolution multiobject spectrographs include HERMES on AAT (spectral resolution $R \sim 28,000$), APOGEE with SDSS ($R \sim 20,000$ in infrared), GAIA-ESO with GIRAFFE and UVES on VLT ($R \sim 20,000/40,000$), WEAVE on WHT, 4MOST on VISATA/NTT, and the Next-Generation Canada-France-Hawaii Telescope.

To investigate these histories, Freeman & Bland-Hawthorn (2002) have proposed chemical tagging. This approach is based on the assumptions that stars form in clusters that have uniform and unique chemical compositions and that disruptions are a result of the action of mass loss from stellar evolution, two-body effects, and the tidal field of the Galaxy. The goal of chemical tagging is to use element abundances to reconstruct the ancient clusters in which stars were born. Although an interesting approach to study the formation of the Galaxy, chemical tagging may make reconstructing clusters from element abundances difficult if the chemical composition of clusters

does not vary greatly. The variation of stellar yields is caused by differences among progenitor mass, metallicity, energy, and, possibly, mixing and fallback.

Principal component analysis (PCA) of chemical abundances (Andrews et al. 2012, Ting et al. 2012) is an alternative method to disentangle chemical enrichment sources from observed elemental abundances. PCA does not assume a star-formation history, so it is a complementary approach to GCE. However, it is not easy to interpret high-order components; thus, high-quality data are required.

Galactic archaeology surveys will also uncover a number of EMP stars. The metal-poor end of MDFs is important to constrain the early stages of GCE (Section 7). Elemental abundance patterns of EMP stars can constrain the explosion mechanisms of core-collapse supernovae (Section 8). There is a degeneracy in these two constraints, and faint supernovae result in low chemical enrichment efficiency. Therefore, it is necessary to obtain the abundance ratios of many elements, particularly Zn and Mn, to minimize the uncertainties of abundance profiling.

10. FUTURE OUTLOOK

The stellar yields discussed in this review are largely based on spherical (1D) models of the evolution and explosion of massive stars. In some stars, however, the aspherical effect of rotation may be large enough to induce meridional circulation and shear instabilities. The resulting large-scale mixing would then affect stellar structure and yields. Multidimensional calculations of convection also produce significant differences from 1D approximations (Arnett & Meakin 2011). Furthermore, although magnetic fields could have a large effect on angular momentum transport, the treatment of magnetic fields is poorly understood. To account for these processes, a multidimensional implicit stellar evolution code is necessary, but obtaining such a code remains a significant challenge (e.g., Eriguchi & Müller 1991, Kifonidis & Müller 2012). Thus, the effects of rotation have been only approximately taken into account in spherical models with certain parameters (e.g., Heger, Woosley & Spruit 2005; Ekström et al. 2008, 2012; Yoon, Dierks & Langer 2012). Chieffi & Limongi (2013) provide yields from these quasi-aspherical models. Observed abundance ratios of certain elements, e.g., N/C and N/O, as a function of the rotation velocity and metallicity could also provide some useful constraints on such an approach (e.g., Langer 2012).

The explosion mechanism of core-collapse supernovae remains a topic of debate (e.g., Janka 2012, Kotake et al. 2012, Bruenn et al. 2013, Burrows 2013). For $M > 10 M_{\odot}$, 1D models do not give rise to explosions (e.g., Sumiyoshi et al. 2005). The explosion must be aspherical, and 3D convection and/or standing accretion shock instability may play an important role. Explosion energies, ranging from faint supernovae to hypernovae, may depend on the degree of asphericity as well as progenitor mass. Such an aspherical structure could affect nucleosynthesis (Section 8.6). For example, underproduction of Sc, Ti, and V can be improved in high-entropy ejecta of multidimensional models (Maeda & Nomoto 2003, Tominaga et al. 2007, Tominaga 2009) (see **Figure 14**). Eventually, nucleosynthesis simulations based on successful multidimensional explosion models will be necessary, but obtaining such calculations is still quite a challenge. Thus, 1D models with parameters constrained from observations are useful for calculating nucleosynthesis yields for a wide range of parameter spaces. In GCE, the most important parameters of core-collapse supernovae are explosion energy as well as mixing and fallback. Contributions from faint supernovae may be important during the early stages of chemical enrichment (Kobayashi, Tominaga & Nomoto 2011). The neutrino process is also important to account for the underproduction of F, Sc, K, and V, which would be enhanced in nucleosynthesis in a p -rich environment produced by neutrino interactions (Iwamoto et al. 2006, Heger & Woosley 2010, Kobayashi et al. 2011).

Nucleosynthesis in hot bubbles or accretion disks around black holes has been suggested as the origin of Sc, Co, and Zn as well as heavier elements (e.g., Pruet et al. 2005, Fröhlich et al. 2006) (also see Section 8.2.1). The final composition largely varies in response to trajectory experienced by each mass element. Thus, it is difficult to constrain the yield integrated over the ejected materials (Tominaga, Umeda & Nomoto 2007). Such nucleosynthesis depends more strongly on the explosion mechanism and neutrino transport rather than on the explosive nucleosynthesis discussed in this review.

The sites of *s*-processes have been identified as (*a*) the He-shell flashes in AGB and super AGB stars (main *s*-processes) and (*b*) core He burning and shell C burning in massive stars (weak *s*-processes). Yields are still affected by large uncertainties of nuclear reaction rates (for a review, see Käppeler et al. 2011). The physics of assumed convective mixing during He-shell flashes needs to be studied further and should include multidimensional simulations.

The sites of *r*-processes have not been identified (e.g., Qian & Wasserburg 2007), although the following provide suggestions for further investigation (e.g., Thielemann et al. 2011): (*a*) Weak *r*-processes could occur if the neutrino-driven neutron star wind changes from being proton rich to being neutron rich in the later phase. (*b*) Main (strong) *r*-processes require a very neutron-rich environment, which could exist in neutron star mergers or in accretion disks around black holes. Further studies and comparison with GCE models are necessary. PCA is also useful: For example, Ting et al. (2012) showed that production of *r*-process elements is associated with production of α and Fe-peak elements.

Type Ia supernovae are important sources of Fe in the Universe, such that identification of their progenitors is crucial to construct cosmic chemical evolution models as discussed in Section 6. Comparison between GCE models and elemental abundances in stars can place constraints on the progenitor scenario. In particular, production and distribution of ^{58}Ni in Type Ia supernovae strongly constrain the progenitor scenario as well as flame propagation (Nomoto, Kamiya & Nakasato 2013). Overproduction of ^{58}Ni (Thielemann, Nomoto & Yokoi 1986; Kobayashi et al. 2006) is improved in multidimensional models of flame propagation (e.g., Maeda et al. 2010a, Röpke et al. 2012, Seitenzahl et al. 2013), although its distribution in current multidimensional models is not consistent with the nebular spectra of Type Ia supernovae (Section 6.1).

A complete set of the nucleosynthesis yields of various enrichment sources is not available. Nevertheless, such sources may be important to explain the certain elements/isotopes because an offset is seen in comparisons between observations and GCE models. These enrichment sources include the following:

1. Novae for ^{15}N production (Romano & Matteucci 2003).
2. Super AGB stars for production of N and ^{13}C at low metallicities (Kobayashi, Karakas & Umeda 2011) and of ^{48}Ca (Wanajo, Janka & Müller 2013).
3. Some low-metallicity AGB stars if they explode as Type 1.5 supernova; $[\alpha/\text{Fe}]$ ratios are dramatically decreased, which is not seen in observations. If a star has low $[\alpha/\text{Fe}]$ at low $[\text{Fe}/\text{H}]$, it is difficult to distinguish the effect from the contribution of “prompt” Type Ia supernovae because of their similar abundance patterns.
4. 10–13- $M_{\odot}$ stars; their final fates need to be studied further, although supernovae arising from this mass range may not contribute much to GCE.

Even after taking the above sources into account, problems remain. Compared with Big Bang nucleosynthesis, ^7Li and ^6Li are under- and overabundant, respectively (Asplund et al. 2006). The origin of the O-Na anticorrelation in GCs is also unknown: It may be either AGB stars or rotating massive stars (Section 7.5).

In most GCE models, the following processes are not included: inhomogeneity of the ISM, cosmic rays [which should affect the abundances of Be and B (Prantzos 2012)], and binaries. To include them precisely, more-realistic models that employ hydrodynamical simulations (e.g., Kobayashi & Nakasato 2011) are necessary. Today, models are sufficiently sophisticated, such that they can incorporate new information on chemical yields far better than previous models could. Current models can also make interesting testable predictions about the nature of stellar populations in the Milky Way and Andromeda. Finally, some uncertainties exist regarding important nuclear reaction rates. The experimental and theoretical status of these nuclear reaction rates has been summarized by Wiescher, Käppeler & Langanke (2012).

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

This research has been supported by the World Premier International Research Center Initiative; MEXT, Japan; the Center for Computational Astrophysics of the National Astronomical Observatory of Japan; and the Australian National University. We would like to thank W. Aoki, T. Beers, A. Bunker, S. Campbell, N. Iwamoto, A. Karakas, M. Limongi, K. Maeda, R. Smith, M. Tanaka, and H. Umeda for fruitful discussions.

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